

Language Models are Unsupervised Multitask Learners

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Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

1. Introduction

Machine learning systems now excel (in expectation) at tasks they are trained for by using a combination of large datasets, high-capacity models, and supervised learning (Krizhevsky et al., 2012) (Sutskever et al., 2014) (Amodei et al., 2016). Yet these systems are brittle and sensitive to slight changes in the data distribution (Recht et al., 2018) and task specification (Kirkpatrick et al., 2017). Current systems are better characterized as narrow experts rather than

competent generalists. We would like to move towards more general systems which can perform many tasks – eventually without the need to manually create and label a training dataset for each one.

The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al., 2019) and the two most ambitious efforts to date have trained on a total of 10 and 17 (dataset, objective) pairs respectively (McCann et al., 2018) (Bowman et al., 2018). From a meta-learning perspective, each (dataset, objective) pair is a single training example sampled from the distribution of datasets and objectives. Current ML systems need hundreds to thousands of examples to induce functions which generalize well. This suggests that multitask training many need just as many effective training pairs to realize its promise with current approaches. It will be very difficult to continue to scale the creation of datasets and the design of objectives to the degree that may be required to brute force our way there with current techniques. This motivates exploring additional setups for performing multitask learning.

The current best performing systems on language tasks

语言模型属于无监督多任务学习器。

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摘要

诸如问答、机器翻译、阅读理解以及摘要生成任务等自然语言处理任务，通常是通过在针对特定任务的数据库集上采用监督学习方法来处理的。我们发现，当语言模型在名为WebText的包含数百万网页的新数据集上接受训练时，即便没有任何显式的监督，也能开始学习这些任务。在给定文档及相应问题的条件下，该语言模型生成的答案在CoQA数据集上的F1分数可达55，这一表现与4个基准系统中的3个相当甚至更优，而且全程并未使用那127,000+ 个训练示例。语言模型的容量对于实现零样本任务迁移至关重要，提升其容量能够以对数线性方式提高其在各类任务上的性能。我们规模最大的模型GPT-2是一款拥有15亿参数的Transformer架构模型，在零样本场景下于8个经过测试的语言建模数据集集中的7个取得了当前最佳水平的结果，但在处理WebText数据时仍存在欠拟合现象。该模型的输出样本体现了这些性能提升，其中包含了结构连贯的文本段落。这些研究结果为构建能够从自然出现的实例中学习并执行任务的语言处理系统指明了一条极具前景的发展路径。

1. 引言

目前，机器学习系统借助大规模数据集、高容量模型以及监督学习等手段，（在预期中）已能在其被训练执行的任务上表现出色（克里兹夫斯基等人，2012年）（苏茨克维尔等人，2014年）（阿莫代伊等人，2016年）。然而，这类系统十分脆弱，对数据分布的细微变化以及任务要求的微小改动都极为敏感（雷希特等人，2018年）和任务定义的调整（柯克帕特里克等人，2017年）。实际上，当前的这些系统更应被视为狭域专家系统，而非

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通用型专家系统。我们希望朝着能够处理更多任务的更为通用的系统发展——最终甚至无需为每项任务手动创建并标注训练数据集。

构建机器学习系统的主流方法是为特定任务收集能够展现正确行为模式的训练示例数据集，训练系统模仿这些行为模式，随后在独立同分布的测试样本上评估其性能。这种方法在推动狭域专家系统的发展方面发挥了重要作用。然而，文本描述模型（Lake等人，2017年）、阅读理解系统（Jia与Liang，2017年）以及图像分类器（Alcorn等人，2018年）在面对各种不同类型的输入时常常表现出不稳定的特性，这恰恰凸显出该方法的不足之处。

我们认为，当前系统普遍采用单一领域数据集进行单任务训练，这是导致其泛化能力不足的重要原因。若要利用现有架构打造更为鲁棒的系统，就必须在多种不同的领域和任务上开展训练并评估性能。近期，已有GLUE（Wanget等人，2018年）和decaNLP（McCann等人，2018年）等若干基准测试被提出，用于开展相关研究。

多任务学习（Caruana，1997年）是一种有望提升整体性能的框架。不过，在自然语言处理领域，多任务学习仍然处于起步阶段。近期的一些研究仅报告了较为有限的性能提升（Yogatama等人，2019年），而迄今为止最具有雄心的两项研究分别仅使用了10组和17组(数据集，目标)训练样本（McCann等人，2018年；Bowman等人，2018年）。从元学习的视角来看，每一组(数据集，目标)训练样本实际上都是从各类数据集与目标的分布中抽取出的单个示例。目前的机器学习系统往往需要数百乃至数千个示例才能构建出具备良好泛化能力的模型。由此可见，若要借助现有方法实现多任务学习的潜力，可能同样需要同等数量的有效训练样本。继续以当前的技术手段大规模扩充数据集规模并设计更多目标，恐怕十分困难，这也促使人们开始探索其他可用于开展多任务学习的方案。

目前在语言处理任务中表现最佳的系统

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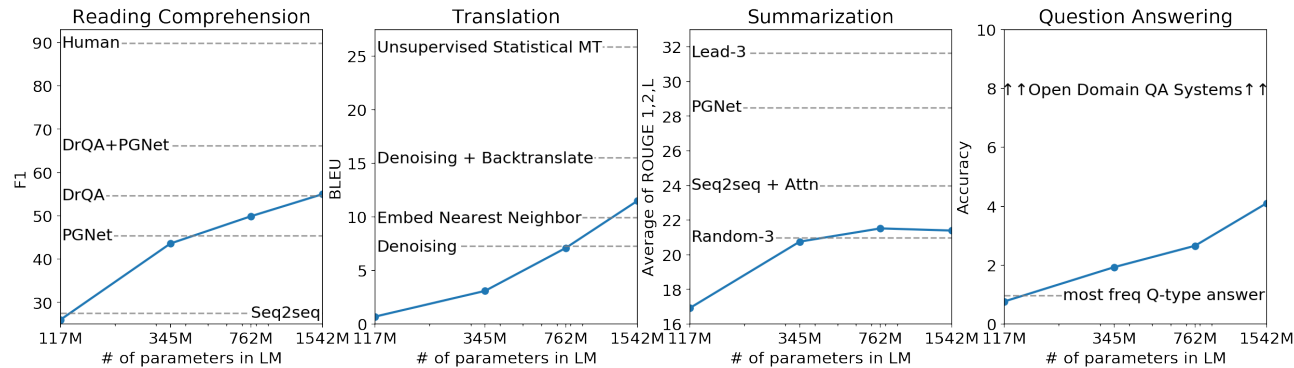


Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

utilize a combination of pre-training and supervised fine-tuning. This approach has a long history with a trend towards more flexible forms of transfer. First, word vectors were learned and used as inputs to task-specific architectures (Mikolov et al., 2013) (Collobert et al., 2011), then the contextual representations of recurrent networks were transferred (Dai & Le, 2015) (Peters et al., 2018), and recent work suggests that task-specific architectures are no longer necessary and transferring many self-attention blocks is sufficient (Radford et al., 2018) (Devlin et al., 2018).

These methods still require supervised training in order to perform a task. When only minimal or no supervised data is available, another line of work has demonstrated the promise of language models to perform specific tasks, such as commonsense reasoning (Schwartz et al., 2017) and sentiment analysis (Radford et al., 2017).

In this paper, we connect these two lines of work and continue the trend of more general methods of transfer. We demonstrate language models can perform down-stream tasks in a zero-shot setting – without any parameter or architecture modification. We demonstrate this approach shows potential by highlighting the ability of language models to perform a wide range of tasks in a zero-shot setting. We achieve promising, competitive, and state of the art results depending on the task.

2. Approach

At the core of our approach is language modeling. Language modeling is usually framed as unsupervised distribution estimation from a set of examples $(x_1, x_2, \dots, x_n)$ each composed of variable length sequences of symbols $(s_1, s_2, \dots, s_n)$. Since language has a natural sequential ordering, it is common to factorize the joint probabilities over

symbols as the product of conditional probabilities (Jelinek & Mercer, 1980) (Bengio et al., 2003):

$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{i-1}) \quad (1)$$

This approach allows for tractable sampling from and estimation of $p(x)$ as well as any conditionals of the form $p(s_{n-k}, \dots, s_n | s_1, \dots, s_{n-k-1})$. In recent years, there have been significant improvements in the expressiveness of models that can compute these conditional probabilities, such as self-attention architectures like the Transformer (Vaswani et al., 2017).

Learning to perform a single task can be expressed in a probabilistic framework as estimating a conditional distribution $p(output | input)$. Since a general system should be able to perform many different tasks, even for the same input, it should condition not only on the input but also on the task to be performed. That is, it should model $p(output | input, task)$. This has been variously formalized in multitask and meta-learning settings. Task conditioning is often implemented at an architectural level, such as the task specific encoders and decoders in (Kaiser et al., 2017) or at an algorithmic level such as the inner and outer loop optimization framework of MAML (Finn et al., 2017). But as exemplified in McCann et al. (2018), language provides a flexible way to specify tasks, inputs, and outputs all as a sequence of symbols. For example, a translation training example can be written as the sequence (translate to french, english text, french text). Likewise, a reading comprehension training example can be written as (answer the question, document, question, answer). McCann et al. (2018) demonstrated it was possible to train a single model, the MQAN,

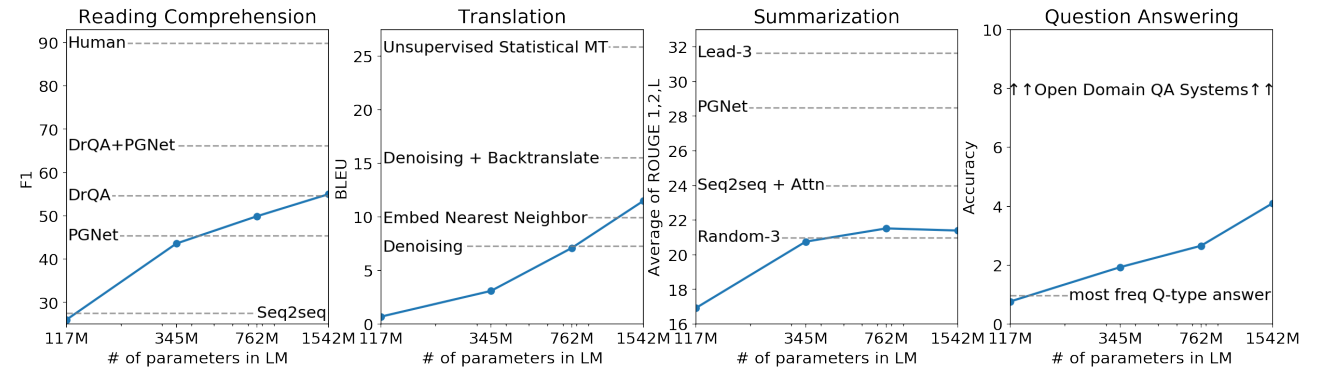


图1。不同模型规模下WebText语言模型在多种自然语言处理任务中的零样本场景性能表现。其中阅读理解任务的测试数据来自CoQA数据集（Reddy等人，2018年）；翻译任务的测试数据来自WMT-14法语到英语竞赛（Artetxe等人，2017年）；摘要生成任务的测试数据来自CNN和《每日邮报》相关数据集（See等人，2017年）；问答任务则基于自然问题数据集进行测试（Kwiatkowski等人，2019年）。第3章将对各项测试结果进行详细说明。

这类系统会结合预训练与监督微调技术。该方法有着悠久的历史，且呈现出向更灵活的迁移方式发展的趋势。最初，人们先学习词向量并将其作为输入用于任务专用架构（Mikolov等人，2013年）（Collobert等人，2011年），随后又出现了循环神经网络的上下文表示迁移技术（Dai与Le，2015年）（Peters等人，2018年）。而近期的研究则表明，任务专用架构已不再必不可少，直接迁移多个自注意力块便已足够（Radford等人，2018年）（Devlin等人，2018年）。

这些方法在执行任务时仍需要依赖有监督训练。而当可用既有监督数据极少或根本不存在时，另一类研究则展现了语言模型在处理特定任务方面的潜力，比如常识推理（Schwartz等人，2017年）以及情感分析（Radford等人，2017年）。

在本文中，我们将这两类研究思路相结合，延续了向更通用的迁移学习方法发展的趋势。我们证明了语言模型无需进行任何参数调整或架构修改，便能在零样本场景下完成下游任务。通过展示语言模型在零样本场景下可处理多种任务的能力，我们证实了这种方法的巨大潜力，并且根据不同的任务类型，取得了具有竞争力且处于当前最佳水平的结果。

2. 方法

我们方法的核心在于语言建模。语言建模通常被视为一项基于一组示例 $(x_1, x_2, \dots, x_n)$ 的无监督分布估计任务，而这些示例各自又由长度可变的符号序列 $(s_1, s_2, \dots, s_n)$ 构成。由于语言本身具有天然的顺序结构，人们常常将相关的条件概率的联合概率分解为多个独立的部分。

符号的条件概率的乘积形式（Jelinek & Mercer, 1980年）（Bengio等人，2003年）：

$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{i-1}) \quad (1)$$

该方法能够实现高效且易于处理的采样与估计 $p(x)$ ，同时也可处理任何如下形式的条件表达式 $p(s_{n-k}, \dots, s_n | s_1, \dots, s_{n-k-1})$ 。近年来，那些具备计算此类条件概率能力的模型的表达能力取得了显著提升，典型的代表便是采用自注意力机制的Transformer架构（Vaswani等人，2017年）。

在概率框架下，学习执行单一任务可以被表述为对条件分布的估计 $p(output | input)$ 。由于通用系统应当能够处理众多不同的任务，即便面对相同的输入，它也不仅需要依据输入本身，还需结合所需执行的任务来进行判断。也就是说，它应当对 $p(output | input, task)$ 进行建模。这一理念已在多任务学习与元学习领域被以多种形式加以形式化。任务条件化通常在架构层面实现，例如（Kaiser等人，2017年）所提出的针对特定任务的编码器与解码器；也可通过算法层面来实现，比如（Finn等人，2017年）提出的MAML算法中的内外循环优化框架。但正如（McCann等人(2018年)所示，语言提供了一种灵活的方式，能够将任务、输入与输出均以符号序列的形式来指定。例如，一个翻译训练示例就可以被写成如下序列(翻译成法语，英文文本，法文文本)。同样地，阅读理解训练示例也可以表示为(回答问题，文档，问题，答案)。McCann等人（2018年）证明了确实可以通过训练一个名为MQAN的模型，

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to infer and perform many different tasks on examples with this type of format. Language modeling is also able to, in principle, learn the tasks of McCann et al. (2018) without the need for explicit supervision of which symbols are the outputs to be predicted. Since the supervised objective is the the same as the unsupervised objective but only evaluated on a subset of the sequence, the global minimum of the unsupervised objective is also the global minimum of the supervised objective. In this slightly toy setting, the concerns with density estimation as a principled training objective discussed in (Sutskever et al., 2015) are side stepped. The problem instead becomes whether we are able to, in practice, optimize the unsupervised objective to convergence. Preliminary experiments confirmed that sufficiently large language models are able to perform multitask learning in this toy-ish setup but learning is much slower than in explicitly supervised approaches. While it is a large step from the well-posed setup described above to the messiness of “language in the wild”, Weston (2016) argues, in the context of dialog, for the need to develop systems capable of learning from natural language directly and demonstrated a proof of concept – learning a QA task without a reward signal by using forward prediction of a teacher’s outputs. While dialog is an attractive approach, we worry it is overly restrictive. The internet contains a vast amount of information that is passively available without the need for interactive communication. Our speculation is that a language model with sufficient capacity will begin to learn to infer and perform the tasks demonstrated in natural language sequences in order to better predict them, regardless of their method of procurement. If a language model is able to do this it will be, in effect, performing unsupervised multitask learning. We test whether this is the case by analyzing the performance of language models in a zero-shot setting on a wide variety of tasks.
”I’m not the cleverest man in the world, but like they say in French: Je ne suis pas un imbécile [I’m not a fool]. In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: ”Mentez mentez, il en restera toujours quelque chose,” which translates as, ”Lie lie and something will always remain.” “I hate the word ‘perfume,”’ Burr says. ‘It’s somewhat better in French: ‘parfum.’ If listened carefully at 29:55, a conversation can be heard between two guys in French: “-Comment on fait pour aller de l’autre coté? -Quel autre coté?”, which means “- How do you get to the other side? - What side?”. If this sounds like a bit of a stretch, consider this question in French: As-tu aller au cinéma?, or Did you go to the movies?, which literally translates as Have-you to go to movies/theater? “Brevet Sans Garantie Du Gouvernement”, translated to English: “Patented without government warranty”.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

Common Crawl. [Trinh & Le \(2018\)](#)’s best results were achieved using a small subsample of Common Crawl which included only documents most similar to their target dataset, the Winograd Schema Challenge. While this is a pragmatic approach to improve performance on a specific task, we want to avoid making assumptions about the tasks to be performed ahead of time.

Instead, we created a new web scrape which emphasizes document quality. To do this we only scraped web pages which have been curated/filtered by humans. Manually filtering a full web scrape would be exceptionally expensive so as a starting point, we scraped all outbound links from Reddit, a social media platform, which received at least 3 karma. This can be thought of as a heuristic indicator for whether other users found the link interesting, educational, or just funny.

The resulting dataset, WebText, contains the text subset of these 45 million links. To extract the text from HTML responses we use a combination of the Dragnet ([Peters & Lecocq, 2013](#)) and Newspaper¹ content extractors. All results presented in this paper use a preliminary version of WebText which does not include links created after Dec 2017 and which after de-duplication and some heuristic based cleaning contains slightly over 8 million documents for a total of 40 GB of text. We removed all Wikipedia documents from WebText since it is a common data source for other datasets and could complicate analysis due to over-

¹<https://github.com/codelucas/newspaper>

语言模型属于无监督多任务学习器。
“我虽算不上世界上最聪明的人，但用法语来说就是：Jenesuispas unimbécile [我不是傻子]。” 在8月16日一篇现已删除的文章中，乔利埃特选区的保守党候选人 Soheil Eid用法语写道：Mentez mentez,ilen restera toujours quelque chose，其含义为：撒谎吧，撒谎吧，总会有些东西留下来。 Burr说道：‘我讨厌“香水”这个词。用法语表达的“parfum”要好听一些。’ 如果在29分55秒处仔细聆听，可以听到两个法国人之间的对话：“-Comment on fait pour aller de l’ autre cot’e? -Quel autre cot’e?”，其意思是：-要怎么去另一边? -是哪一边？ 如果这听起来有些牵强，不妨试想这样一个法语问题：As-tu aller au cinéma?, 或者Did you go tothe movies?，其字面意思其实是‘你去看电影/剧院了吗?’。 “Brevet Sans Garantie Du Gouvernement”，对应的英文翻译为：” Patented without government warranty。”

表1。这些是在WebText训练集中发现的、自然存在的英语到法语以及法语到英语翻译示例。

Common Crawl. [Trinh与Le\(2018年\)](#)的研究团队通过使用仅包含与目标数据集Winograd模式挑战赛最为相似的文档的小型子样本，取得了最佳效果。虽然这是一种提升特定任务性能的实用方法，但我们希望避免提前对未来要处理的任务做出假设。

相反，我们开发了一种新的网页抓取方式，更注重文档的质量。为此，我们只抓取那些经过人类筛选/整理过的网页。若要对全部网页进行人工筛选成本极高，因此我们最初从社交平台Reddit上抓取了所有获得至少3点karma值的对外链接，这类指标可视为其他用户认为该链接有趣、有教育意义或单纯好笑的启发式判断依据。

最终得到的数据集WebText包含了这4500万个链接对应的文本子集。为从HTML响应中提取文本，我们采用了 Dragnet ([Peters &Lecocq、2013](#)) 和Newspaper¹ 这两种文本提取工具的组合。本文中呈现的所有结果均基于WebText的早期版本，该版本未包含2017年12月之后生成的链接；经过去重处理以及一些基于规则的清洗后，该数据集共包含略多于800万份文档，文本总量达40GB。由于维基百科是其他众多数据集的常见来源，且其中的重复内容可能会使分析变得复杂，因此我们已将其所有文档从WebText中剔除。

¹<https://github.com/codelucas/newspaper>

从而利用这类格式的示例来推断并完成各种不同的任务。

从原理上来讲，语言建模同样能够学习[McCann等人（2018年）](#)所提出的那些任务，而且无需通过显式监督来确定哪些符号是需要预测的输出。由于有监督目标与无监督目标本质相同，只是评估数据仅来自序列的子集，因此无监督目标的全局最小值同时也是有监督目标的全局最小值。在这样一个较为简化的场景中，苏茨克维尔等人（[2015年](#)）所探讨的、将密度估计作为核心训练目标的相关问题便不再适用。此时问题转变为：在实际应用中，我们是否能够让无监督目标收敛到最优解。初步实验表明，规模足够大的语言模型确实能够在这样的简化环境中实现多任务学习，但其学习速度远慢于采用显式监督的方法。

尽管从上文描述的理想化设定到现实中“真实语言环境”的复杂状况之间存在巨大差距，[Weston（2016年）](#)在对话系统的研究背景下指出，有必要开发能够直接从自然语言中学习的系统，并通过利用教师输出的前向预测来学习问答任务，以此证明这一构想的可行性——即无需奖励信号即可完成该任务。虽然对话系统是一种颇具吸引力的方法，但我们认为它存在过度限制的缺陷。互联网上存在着大量无需通过交互式沟通即可被动获取的信息。我们的推测是，具备足够处理能力的语言模型将会开始学会推断并执行自然语言序列中展现出的各种任务，从而更精准地预测这些内容，而无论其获取信息的方式为何。如果语言模型真的能够做到这一点，那么实际上它就是在执行无监督多任务学习。我们通过分析语言模型在零样本场景下处理多种不同任务的性能，来验证这一观点是否成立。

2.1. 训练数据集

此前大多数研究都是在单一类型的文本上对语言模型进行训练的，比如新闻文章（[Jozefowicz等人，2016年](#)）、维基百科（[Merity等人，2016年](#)），或是小说书籍（[Kiros等人，2015年](#)）。我们的方法旨在构建尽可能庞大且多样化的数据集，从而收集来自尽可能多不同领域和上下文的自然语言任务示例。

另一类极具潜力、能够提供丰富且几乎不受限制的文本资源，便是诸如Common Crawl这样的网络爬取数据。尽管这些存档的规模比现有的语言建模数据集大出数个数量级，但它们的数据质量存在相当严重的问题。[Trinh与Le\(2018年\)](#)在研究常识推理时使用了Common Crawl，但他们指出其中大量文档“的内容大多难以理解”。我们在最初的实验中也观察到了类似的数据问题

lapping training data with test evaluation tasks.

2.2. Input Representation

A general language model (LM) should be able to compute the probability of (and also generate) any string. Current large scale LMs include pre-processing steps such as lower-casing, tokenization, and out-of-vocabulary tokens which restrict the space of model-able strings. While processing Unicode strings as a sequence of UTF-8 bytes elegantly fulfills this requirement as exemplified in work such as [Gillick et al. \(2015\)](#), current byte-level LMs are not competitive with word-level LMs on large scale datasets such as the One Billion Word Benchmark ([Al-Rfou et al., 2018](#)). We observed a similar performance gap in our own attempts to train standard byte-level LMs on WebText.

Byte Pair Encoding (BPE) ([Sennrich et al., 2015](#)) is a practical middle ground between character and word level language modeling which effectively interpolates between word level inputs for frequent symbol sequences and character level inputs for infrequent symbol sequences. Despite its name, reference BPE implementations often operate on Unicode code points and not byte sequences. These implementations would require including the full space of Unicode symbols in order to model all Unicode strings. This would result in a base vocabulary of over 130,000 before any multi-symbol tokens are added. This is prohibitively large compared to the 32,000 to 64,000 token vocabularies often used with BPE. In contrast, a byte-level version of BPE only requires a base vocabulary of size 256. However, directly applying BPE to the byte sequence results in sub-optimal merges due to BPE using a greedy frequency based heuristic for building the token vocabulary. We observed BPE including many versions of common words like `dog` since they occur in many variations such as `dog.` `dog!` `dog?` . This results in a sub-optimal allocation of limited vocabulary slots and model capacity. To avoid this, we prevent BPE from merging across character categories for any byte sequence. We add an exception for spaces which significantly improves the compression efficiency while adding only minimal fragmentation of words across multiple vocab tokens.

This input representation allows us to combine the empirical benefits of word-level LMs with the generality of byte-level approaches. Since our approach can assign a probability to any Unicode string, this allows us to evaluate our LMs on any dataset regardless of pre-processing, tokenization, or vocab size.

2.3. Model

We use a Transformer ([Vaswani et al., 2017](#)) based architecture for our LMs. The model largely follows the details of the OpenAI GPT model ([Radford et al., 2018](#)) with a

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Table 2. Architecture hyperparameters for the 4 model sizes.

few modifications. Layer normalization ([Ba et al., 2016](#)) was moved to the input of each sub-block, similar to a pre-activation residual network ([He et al., 2016](#)) and an additional layer normalization was added after the final self-attention block. A modified initialization which accounts for the accumulation on the residual path with model depth is used. We scale the weights of residual layers at initialization by a factor of $1/\sqrt{N}$ where N is the number of residual layers. The vocabulary is expanded to 50,257. We also increase the context size from 512 to 1024 tokens and a larger batchsize of 512 is used.

3. Experiments

We trained and benchmarked four LMs with approximately log-uniformly spaced sizes. The architectures are summarized in Table 2. The smallest model is equivalent to the original GPT, and the second smallest equivalent to the largest model from BERT ([Devlin et al., 2018](#)). Our largest model, which we call GPT-2, has over an order of magnitude more parameters than GPT. The learning rate of each model was manually tuned for the best perplexity on a 5% held-out sample of WebText. All models still underfit WebText and held-out perplexity has as of yet improved given more training time.

3.1. Language Modeling

As an initial step towards zero-shot task transfer, we are interested in understanding how WebText LM’s perform at zero-shot domain transfer on the primary task they are trained for – language modeling. Since our model operates on a byte level and does not require lossy pre-processing or tokenization, we can evaluate it on any language model benchmark. Results on language modeling datasets are commonly reported in a quantity which is a scaled or exponentiated version of the average negative log probability per canonical prediction unit - usually a character, a byte, or a word. We evaluate the same quantity by computing the log-probability of a dataset according to a WebText LM and dividing by the number of canonical units. For many of these datasets, WebText LMs would be tested significantly out-of-distribution, having to predict aggressively standardized text, tokenization artifacts such as disconnected punctuation and contractions, shuffled sentences, and even the string

训练数据与测试评估任务之间存在重叠现象。

2.2. 输入表示

一个通用的语言模型应当能够计算任意字符串的概率，并且还能生成这样的字符串。当前的大规模语言模型包含诸如小写转换、分词以及处理词汇表外标记等预处理步骤，而这些步骤会限制可被模型处理的字符串范围。虽然将Unicode字符串视为UTF-8字节序列来处理能够很好地满足这一要求，正如Gillick等人（2015年）的相关研究所示，但目前的基于字节的语言模型在面对“十亿词基准测试”这类大规模数据集时，仍无法与基于单词的语言模型相抗衡——Al-Rfou等人 在2018年的研究也证实了这一点。我们在尝试利用WebText数据集训练标准的基于字节的语言模型时，也观察到了类似的性能差距。Gillick等人（2015年）Al-Rfou等人，2018年

字节对编码（BPE）（[Sennrich等人，2015年](#)）是一种介于字符级和单词级语言建模之间的实用方案，它能够针对出现频率较高的符号序列采用单词级输入，而对于出现频率较低的符号序列则使用字符级输入。尽管其名称中带有“字节”二字，但实际上常见的BPE实现是基于Unicode代码点而非字节序列运行的。若要使用这类实现来建模所有的Unicode字符串，就必须将全部Unicode符号纳入基础词汇表中，这样一来，在添加任何多符号标记之前，基础词汇表的规模就已经会超过13万条。相比BPE通常所使用的3万到6万条标记的词汇表，这个规模实在大得难以接受。相比之下，基于字节的BPE版本只需要256条作为基础词汇表就足够了。不过，如果直接对字节序列应用BPE，由于该算法在构建标记词汇表时会采用基于贪婪频率的启发式方法，因此合并效果往往不尽如人意。我们观察到，BPE会将很多常见单词的不同变体都归为同一类别，比如`dog`，因为这类单词会有“`dog.` `dog!` `dog?`”等诸多不同形式存在。这样一来，有限的词汇表槽位和模型容量就会被分配得不够合理。为了解决这个问题，我们阻止BPE对任意字节序列中的不同字符类别进行合并，同时为空格设置了特殊例外规则，这不仅显著提升了压缩效率，也仅会在少数情况下导致单词在多个词汇表标记之间出现轻微拆分。

这种输入表示方式让我们能够将基于词级的语言模型的实际优势与基于字节级方法的通用性相结合。由于我们的方法能够为任意Unicode字符串赋予概率值，因此无论数据集的预处理、分词方式或词汇表规模如何，我们都能够对相应的语言模型进行评估。

2.3. 模型

我们使用的Vaswani等人 在2017年提出的Transformer架构作为构建我们所用语言模型的基础。该模型在整体设计上参考了Radford等人 在2018年设计的OpenAI GPT模型的诸多细节，同时仅做了少量修改。

参数	层数	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542百万	48	1600

表2. 4种模型尺寸的架构超参数。

层归一化技术Ba等人 在2016年提出后，被应用到了每个子模块的输入端，其运作方式类似预激活残差网络He等人 在2016年提出的那种结构；此外，在最后一个自注意力模块之后还增加了一层层归一化处理。我们还采用了经过改进的初始化方法，该方法能够考虑到随着模型深度增加而积累在残差路径上的数值。在初始化阶段，我们会按照 $1/\sqrt{N}$ 的比例对残差层的权重进行缩放，一其中 N 指的是残差层的总数量。与此同时，该语言模型的词汇表规模被扩大到50,257个元素，上下文长度也从512个标记提升至1024个标记，批量大小也相应增大到了512。

3. 实验

我们训练了四种参数规模大致遵循对数均匀分布的LM，并对其进行了基准测试。这些LM的架构总结在表2中。规模最小的模型相当于最初的GPT，第二小的则相当于BERT中的最大模型（[Devlin等人，2018年](#)）。我们称之为GPT-2的最大模型，其参数数量比GPT多出一个数量级以上。为了在WebText数据集抽取的5%测试样本上获得最低的困惑度，我们手动调整了每种模型的学习率。尽管如此，所有模型依然存在欠拟合现象，而且随着训练时间的增加，测试集上的困惑度目前仍未出现下降。

3.1. 语言建模

作为实现零样本任务迁移的第一步，我们希望了解WebText语言模型在处理其最初被训练用于的语言建模这一核心任务时的零样本领域自适应表现如何。由于该模型在字节层面运行，且无需经过有损的预处理或分词操作，因此我们可以在任何语言模型基准测试上对其性能进行评估。在语言建模数据集上的评估结果通常以每个标准预测单元的平均负对数概率的缩放值或指数形式来呈现——这类标准预测单元通常是字符、字节或单词。我们的评估方法也是类似，即先根据WebText语言模型计算出数据集的对数概率，再除以标准预测单元的数量。对于许多此类数据集而言，WebText语言模型实际上是在完全偏离原始数据分布的情况下进行测试的，它们必须预测经过严格标准化的文本、各种分词伪影如断裂的标点符号和缩写形式、被打乱顺序的句子，甚至是特定的字符串。

Language Models are Unsupervised Multitask Learners										
	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

Table 3. Zero-shot results on many datasets. No training or fine-tuning was performed for any of these results. PTB and WikiText-2 results are from (Gong et al., 2018). CBT results are from (Bajgar et al., 2016). LAMBADA accuracy result is from (Hoang et al., 2018) and LAMBADA perplexity result is from (Grave et al., 2016). Other results are from (Dai et al., 2019).

<UNK> which is extremely rare in WebText - occurring only 26 times in 40 billion bytes. We report our main results in Table 3 using invertible de-tokenizers which remove as many of these tokenization / pre-processing artifacts as possible. Since these de-tokenizers are invertible, we can still calculate the log probability of a dataset and they can be thought of as a simple form of domain adaptation. We observe gains of 2.5 to 5 perplexity for GPT-2 with these de-tokenizers.

WebText LMs transfer well across domains and datasets, improving the state of the art on 7 out of the 8 datasets in a zero-shot setting. Large improvements are noticed on small datasets such as Penn Treebank and WikiText-2 which have only 1 to 2 million training tokens. Large improvements are also noticed on datasets created to measure long-term dependencies like LAMBADA (Paperno et al., 2016) and the Children’s Book Test (Hill et al., 2015). Our model is still significantly worse than prior work on the One Billion Word Benchmark (Chelba et al., 2013). This is likely due to a combination of it being both the largest dataset and having some of the most destructive pre-processing - 1BW’s sentence level shuffling removes all long-range structure.

3.2. Children’s Book Test

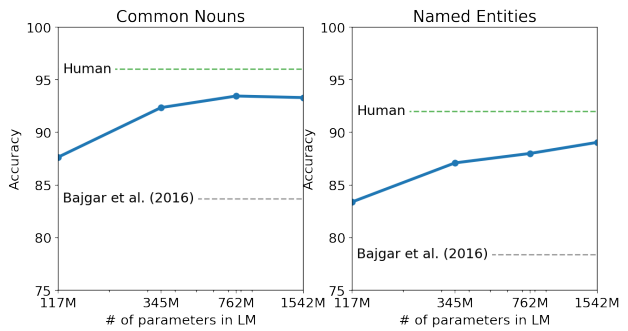


Figure 2. Performance on the Children’s Book Test as a function of model capacity. Human performance are from Bajgar et al. (2016), instead of the much lower estimates from the original paper.

The Children’s Book Test (CBT) (Hill et al., 2015) was created to examine the performance of LMs on different categories of words: named entities, nouns, verbs, and prepositions. Rather than reporting perplexity as an evaluation metric, CBT reports accuracy on an automatically constructed cloze test where the task is to predict which of 10 possible choices for an omitted word is correct. Following the LM approach introduced in the original paper, we compute the probability of each choice and the rest of the sentence conditioned on this choice according to the LM, and predict the one with the highest probability. As seen in Figure 2 performance steadily improves as model size is increased and closes the majority of the gap to human performance on this test. Data overlap analysis showed one of the CBT test set books, The Jungle Book by Rudyard Kipling, is in WebText, so we report results on the validation set which has no significant overlap. GPT-2 achieves new state of the art results of 93.3% on common nouns and 89.1% on named entities. A de-tokenizer was applied to remove PTB style tokenization artifacts from CBT.

3.3. LAMBADA

The LAMBADA dataset (Paperno et al., 2016) tests the ability of systems to model long-range dependencies in text. The task is to predict the final word of sentences which require at least 50 tokens of context for a human to successfully predict. GPT-2 improves the state of the art from 99.8 (Grave et al., 2016) to 8.6 perplexity and increases the accuracy of LMs on this test from 19% (Dehghani et al., 2018) to 52.66%. Investigating GPT-2’s errors showed most predictions are valid continuations of the sentence, but are not valid final words. This suggests that the LM is not using the additional useful constraint that the word must be the final of the sentence. Adding a stop-word filter as an approximation to this further increases accuracy to 63.24%, improving the overall state of the art on this task by 4%. The previous state of the art (Hoang et al., 2018) used a different restricted prediction setting where the outputs of the model were constrained to only words that appeared in the context. For GPT-2, this restriction is harmful rather than helpful

语言模型属于无监督多任务学习器。										
	LAMBADA数据集 (PPL)	LAMBADA数据集 (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 text8数据集 (BPB) (BPC)		WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542百万	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

表3。在多个数据集上的零样本测试结果。这些结果均未经过任何训练或微调处理。PTB和WikiText-2的相关结果出自（Gong等人，2018年）。CBT的结果出自（Bajgar等人，2016年）。LAMBADA的准确率结果出自（Hoang等人，2018年），而LAMBADA的困惑度结果则出自（Grave等人，2016年）。其余结果则来自（Dai等人，2019年）。

<UNK> 这类字符在WebText数据集中极为罕见——在400亿字节的数据中仅出现26次。我们利用可逆反分词器尽可能消除这些分词或预处理过程中产生的伪影，然后在表3中展示了主要实验结果。由于这些反分词器具备可逆性，我们依然能够计算数据集的对数概率，而且它们也可以被视为一种简单形式的领域自适应技术。实验结果显示，使用这类反分词器后，GPT-2的困惑度降低了2.5到5个单位。

WebText语言模型能够在不同领域和数据集之间实现良好的迁移能力，在零样本场景下8个数据集中有7个的准确率都达到了当前最佳水平。在像Penn Treebank和WikiText-2这类仅有100万到200万训练标记的小型数据集上，其性能也有了显著提升；而在那些专为衡量长期依赖关系而设计的数据集，如LAMBADA数据集（Paperno等人，2016年）以及儿童书籍测试数据集（Hill等人，2015年），同样观察到了明显的性能进步。不过在我们的模型在十亿词基测试上的表现仍远远落后于以往的研究成果（Chelba等人，2013年）。这很可能是由于该测试数据集规模最大，同时还存在一些极具破坏性的预处理方式——1BW数据集采用的句子级随机打乱操作会破坏所有长距离结构。

3.2. 儿童书籍测试

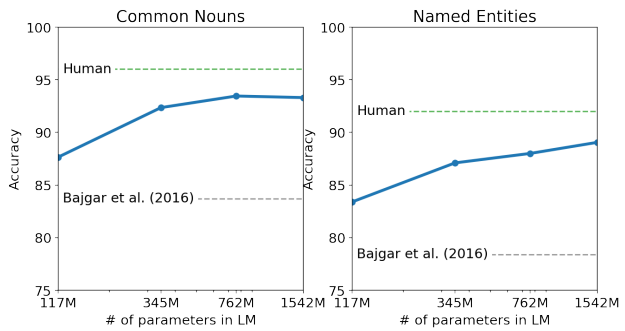


图2。模型容量不同情况下在儿童书籍测试中的表现。此处列出的人类表现数据来源于Bajgar等人（2016年），而非原始论文中给出的那些更低数值。

儿童书籍测试（CBT）（Hill等人，2015年）旨在评估语言模型在各类词汇上的表现，这些词汇包括命名实体、名词、动词和介词。与通常使用困惑度作为评估指标不同，CBT通过一种自动构建的完形测试来衡量准确率，测试要求参与者从10个可能的选项中选出被省略单词的正确答案。我们沿用了原始论文中提出的语言模型处理方法，先计算每个选项在给定该选项的前提下整个句子的概率，再预测概率最高的那个选项。如图2所示，随着模型规模的增大，其性能会持续提升，而且与人类在该测试中的表现之间的差距也在不断缩小。数据重叠分析显示，CBT测试集所使用的书籍之一——鲁德亚德·吉卜林所著的《丛林之书》也包含在WebText数据集中，因此我们最终使用没有显著数据重叠的验证集来展示结果。GPT-2在该测试中的常见名词识别准确率达到了93.3%，命名实体识别准确率为89.1%，从而创造了新的当前最佳成绩。此外，我们还使用了反分词工具来消除CBT数据中因PTB分词风格而产生的分词伪影。

3.3. LAMBADA

LAMBADA数据集（Paperno等人，2016年）用于测试各种系统对文本中长距离依赖关系的建模能力。该任务的目的是预测句子的最后一个单词，而人类要成功完成这一预测至少需要50个标记的上下文信息。GPT-2将此任务上的当前最佳水平从99.8（Grave等人，2016年）对应的困惑度降低到了8.6，同时也将语言模型在该任务上的准确率从19%（Dehghani等人，2018年）提升到了52.66%。通过对GPT-2预测结果的分析发现，虽然大多数预测内容都是句子的合理延续，但并不符合句子最后一个单词的要求，这表明该语言模型并未充分利用“单词必须是句子结尾”这一重要约束条件。通过加入停用词过滤器来近似实现这一约束，准确率进一步提升至63.24%，使得该任务的整体最佳水平又提升了4%。此前的最佳成果（Hoang等人，2018年）采用的是另一种受限预测方式，即仅允许模型输出出现在上下文中的单词；但对GPT-2而言，这种限制非但无益反而有害

since 19% of answers are not in context. We use a version of the dataset without preprocessing.

3.4. Winograd Schema Challenge

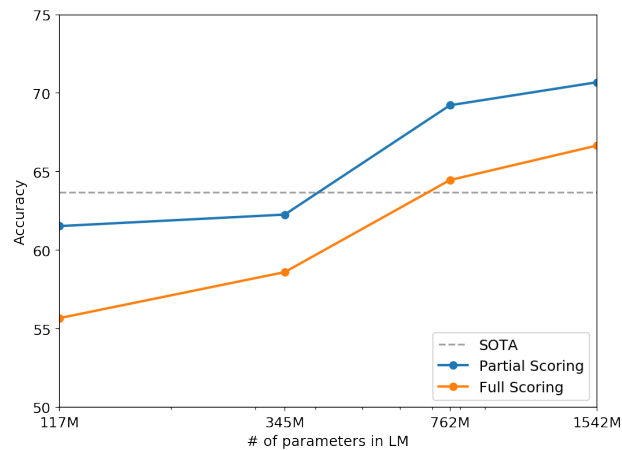


Figure 3. Performance on the Winograd Schema Challenge as a function of model capacity.

The Winograd Schema challenge (Levesque et al., 2012) was constructed to measure the capability of a system to perform commonsense reasoning by measuring its ability to resolve ambiguities in text. Recently Trinh & Le (2018) demonstrated significant progress on this challenge using LMs, by predicting the resolution of the ambiguity with higher probability. We follow their problem formulation and visualize the performance of our models with both full and partial scoring techniques in Figure 3. GPT-2 improves state of the art accuracy by 7%, achieving 70.70%. The dataset is quite small with only 273 examples so we recommend reading Trichelair et al. (2018) to help contextualize this result.

3.5. Reading Comprehension

The Conversation Question Answering dataset (CoQA) Reddy et al. (2018) consists of documents from 7 different domains paired with natural language dialogues between a question asker and a question answerer about the document. CoQA tests reading comprehension capabilities and also the ability of models to answer questions that depend on conversation history (such as “Why?”).

Greedy decoding from GPT-2 when conditioned on a document, the history of the associated conversation, and a final token A: achieves 55 F1 on the development set. This matches or exceeds the performance of 3 out of 4 baseline systems without using the 127,000+ manually collected question answer pairs those baselines were trained on. The supervised SOTA, a BERT based system (Devlin et al.,

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attn	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2 no hint	21.58	4.03	19.47	15.03

Table 4. Summarization performance as measured by ROUGE F1 metrics on the CNN and Daily Mail dataset. Bottom-Up Sum is the SOTA model from (Gehrmann et al., 2018)

2018), is nearing the 89 F1 performance of humans. While GPT-2’s performance is exciting for a system without any supervised training, some inspection of its answers and errors suggests GPT-2 often uses simple retrieval based heuristics such as *answer with a name from the document in response to a who question*.

3.6. Summarization

We test GPT-2’s ability to perform summarization on the CNN and Daily Mail dataset (Nallapati et al., 2016). To induce summarization behavior we add the text TL;DR: after the article and generate 100 tokens with Top- k random sampling (Fan et al., 2018) with $k = 2$ which reduces repetition and encourages more abstractive summaries than greedy decoding. We use the first 3 generated sentences in these 100 tokens as the summary. While qualitatively the generations resemble summaries, as shown in Table 14, they often focus on recent content from the article or confuse specific details such as how many cars were involved in a crash or whether a logo was on a hat or shirt. On the commonly reported ROUGE 1,2,L metrics the generated summaries only begin to approach the performance of classic neural baselines and just barely outperforms selecting 3 random sentences from the article. GPT-2’s performance drops by 6.4 points on the aggregate metric when the task hint is removed which demonstrates the ability to invoke task specific behavior in a language model with natural language.

3.7. Translation

We test whether GPT-2 has begun to learn how to translate from one language to another. In order to help it infer that this is the desired task, we condition the language model on a context of example pairs of the format `english sentence = french sentence` and then after a final prompt of `english sentence = we sample from the model with greedy decoding and use the first generated sentence as the translation`. On the WMT-14 English-French test set, GPT-2 gets 5 BLEU, which is slightly worse than a word-by-word substitution with a bilingual lexicon inferred in previous work on unsupervised word translation

因为有19%的答案并不处于上下文中，因此我们直接使用该数据集而无需进行任何预处理。

3.4. Winograd模式挑战赛

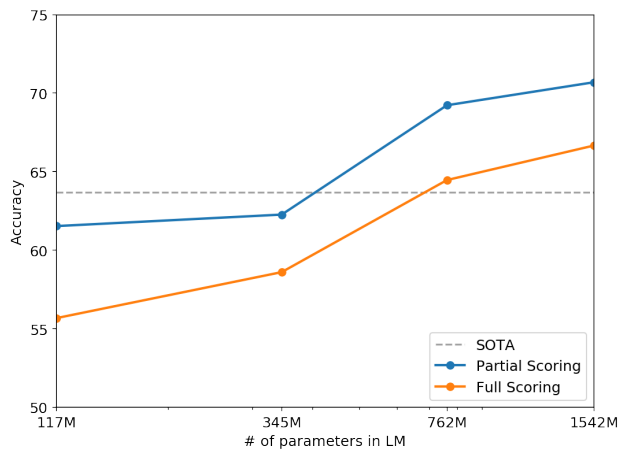


图3。不同模型容量下在Winograd模式挑战赛上的表现。

Winograd模式挑战赛（Levesque等人，2012年）旨在通过检测系统解决文本歧义的能力来衡量其进行常识推理的水平。近期，Trinh与Le(2018年)利用语言模型在该挑战赛中取得了显著进展，他们能够以更高的概率预测出歧义的解决方案。我们沿用了他们的问题设定，并在图3中用完整评分方法和部分评分方法展示了我们模型的性能。GPT-2将当前最佳水平的准确率提升了7%，达到了70.70%。由于该数据集仅包含273个示例，规模相当小，因此我们建议阅读Trichelair等人(2018年)的相关研究，以便更好地理解这一结果背后的背景。

3.5. 阅读理解

Conversation Question Answering数据集（即CoQA数据集）Reddy等人(2018年)包含了来自7个不同领域的文档，以及针对这些文档由提问者和回答者之间进行的自然语言对话。该数据集不仅用于测试阅读理解能力，还用于评估模型回答那些需要结合对话历史才能理解的问题的能力（例如“为什么？”这类问题）。

在结合文档内容、相关对话历史以及最终一个标记作为条件时，GPT-2所采用的贪婪解码A：在开发集上的F1分数可达55。这一成绩甚至超过了4个基准系统中的3个系统的性能，而且无需使用那些用于训练这些基准系统的手动收集的问答对。而当前最佳的监督学习系统则是一种基于BERT模型的系统（Devlin等人。）

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum模型	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq算法 +注意力机制	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2, 无任务提示	21.58	4.03	19.47	15.03

表4.基于CNN与《每日邮报》数据集，通过ROUGE F1指标测得的摘要生成任务性能。Bottom-Up Sum模型为当前最佳性能模型，相关研究出自（格赫尔曼等人，2018年）

2018年），其F1分数已接近人类水平的89分。尽管对于一个未经任何监督训练的系统而言，GPT-2的表现已经相当出色，但对其答案和错误的具体分析表明，该模型常常会采用一些简单的基于检索的启发式方法，比如在遇到关于“谁”的问题时，直接从文档中选取名称作为答案。

3.6. 摘要生成任务

我们在CNN与《每日邮报》数据集上测试了GPT-2的摘要生成能力（纳拉帕蒂等人，2016年）。为了引导模型产生摘要内容，我们在文章末尾添加了TL;DR:这样的文字，随后使用Top-random sampling方法（范等人，2018年）生成100个标记，该方法能够减少内容重复，从而促使模型生成更具抽象性的摘要，而非仅进行简单的逐字解码。我们以这100个标记中生成的前三句话作为最终摘要。虽然从质量上看这些生成的文本确实类似摘要，但如表14所示，它们往往只关注文章中的近期内容，或者会混淆一些具体细节，比如事故中涉及多少辆车，或是帽子或衬衫上是否有徽标。在常用的ROUGE 1、2、L指标评估下，这些生成摘要的性能仅勉强达到经典神经网络基线模型的水平，甚至略逊于直接从文章中随机选取三句话的做法。当去除任务提示后，GPT-2在该任务上的整体性能下降了6.4分，这一结果充分体现了利用自然语言触发语言模型执行特定任务的能力。

3.7. 翻译任务

我们测试了GPT-2是否已经开始学习如何实现一种语言到另一种语言的翻译。为帮助该语言模型推断出这正是所需完成的翻译任务，我们会给它提供格式为“英文句子 = 法文句子”的示例对作为上下文，随后通过“英文句子 = ”这样的最终提示词，再使用贪婪解码方法从模型中采样，并将生成的第一个句子作为翻译结果。在WMT-14的英法语翻译测试集上，GPT-2的BLEU分数仅为5分，这一成绩略逊于此前在无监督词级翻译研究中所采用的、基于双语词典的逐词替换方法。

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

Table 5. The 30 most confident answers generated by GPT-2 on the development set of Natural Questions sorted by their probability according to GPT-2. None of these questions appear in WebText according to the procedure described in Section 4.

(Conneau et al., 2017b). On the WMT-14 French-English test set, GPT-2 is able to leverage its very strong English language model to perform significantly better, achieving 11.5 BLEU. This outperforms several unsupervised machine translation baselines from (Artetxe et al., 2017) and (Lample et al., 2017) but is still much worse than the 33.5 BLEU of the current best unsupervised machine translation approach (Artetxe et al., 2019). Performance on this task was surprising to us, since we deliberately removed non-English webpages from WebText as a filtering step. In order to confirm this, we ran a byte-level language detector² on WebText which detected only 10MB of data in the French language which is approximately 500x smaller than the monolingual French corpus common in prior unsupervised machine translation research.

3.8. Question Answering

A potential way to test what information is contained within a language model is to evaluate how often it generates the correct answer to factoid-style questions. Previous showcasing of this behavior in neural systems where all information is stored in parameters such as *A Neural Conversational Model* (Vinyals & Le, 2015) reported qualitative results due to the lack of high-quality evaluation datasets. The recently introduced Natural Questions dataset (Kwiatkowski et al.,

2019) is a promising resource to test this more quantitatively. Similar to translation, the context of the language model is seeded with example question answer pairs which helps the model infer the short answer style of the dataset. GPT-2 answers 4.1% of questions correctly when evaluated by the exact match metric commonly used on reading comprehension datasets like SQUAD.³ As a comparison point, the smallest model does not exceed the 1.0% accuracy of an incredibly simple baseline which returns the most common answer for each question type (who, what, where, etc...). GPT-2 answers 5.3 times more questions correctly, suggesting that model capacity has been a major factor in the poor performance of neural systems on this kind of task as of yet. The probability GPT-2 assigns to its generated answers is well calibrated and GPT-2 has an accuracy of 63.1% on the 1% of questions it is most confident in. The 30 most confident answers generated by GPT-2 on development set questions are shown in Table 5. The performance of GPT-2 is still much, much, worse than the 30 to 50% range of open domain question answering systems which hybridize information retrieval with extractive document question answering (Alberti et al., 2019).

³Alec, who previously thought of himself as good at random trivia, answered 17 of 100 randomly sampled examples correctly when tested in the same setting as GPT-2. He actually only got 14 right but he should have gotten those other 3

²<https://github.com/CLD2Owners/cld2>

问题:	生成答案:	正确:	概率:
《物种起源》这本书是谁写的?	查尔斯·达尔文	✓	83.4%
Ubuntu项目的创始人是谁?	马克·舒特尔沃思	✓	82.0%
绿湾包装工队的四分卫是谁?	阿隆·罗杰斯	✓	81.1%
熊猫是哪个国家的国宝?	中国	✓	76.8%
是谁提出了相对论?	阿尔伯特·爱因斯坦	✓	76.4%
第一部《星球大战》电影是何时上映的?	1977	✓	71.4%
瑞典最常见的血型是什么?	A	✗	70.6%
谁被视为精神分析的奠基人?	西格蒙德·弗洛伊德	✓	69.3%
1969年，是谁率先在月球上留下了足迹?	尼尔·阿姆斯特朗	✓	66.8%
英国规模最大的超市连锁店是哪家?	特易购	✓	65.3%
“shalom”在英语中的含义是什么?	和平	✓	64.0%
《孙子兵法》的作者是谁?	孙武	✓	59.6%
美国面积最大的州是哪个?	加利福尼亚州	✗	59.2%
绿藻属于哪种繁殖方式的例子?	孤雌生殖	✗	56.5%
维克拉姆历在哪个国家被正式采用?	印度	✓	55.6%
谁是撰写《独立宣言》的主要责任人?	托马斯·杰斐逊	✓	53.3%
哪個美國州构成了蒙大拿州的西界?	蒙大拿州	✗	52.3%
谁在《权力的游戏》中饰演戴佛斯爵士?	彼德·丁克拉格	✗	52.1%
谁来任命联邦储备系统的主席?	珍妮特·耶伦	✗	51.5%
请说明将一个细胞核分裂成两个遗传上完全相同的细胞核的过程是什么?	有丝分裂	✓	50.7%
在NBA历史上，谁获得了最多的最有价值球员奖?	迈克尔·乔丹	✗	50.2%
与罗马城相关的河流是哪条?	台伯河	✓	48.6%
首位遭到弹劾的总统是谁?	安德鲁·约翰逊	✓	48.3%
2017年时，国土安全部的负责人是谁?	约翰·凯利	✓	47.0%
欧盟的通用货币被称为什么?	Euro	✓	46.8%
在《星球大战》中，皇帝的名字是什么?	帕尔帕廷	✓	46.5%
去射击场射击需要持有枪支许可证吗?	No	✓	46.4%
1859年，是谁提出进化论作为生物发展的基础?	查尔斯·达尔文	✓	45.7%
在俄罗斯发生爆炸的核电站是哪一座?	切尔诺贝利	✓	45.7%
在最初的《终结者》系列中，约翰·康纳这个角色是由谁扮演的?	阿诺德·施瓦辛格	✗	45.2%

表5。根据GPT-2的算法，从自然问题数据集中训练集生成的30个最具确定性的答案已按其概率高低排序；按照相关规范，这些题目均未出现在WebText数据集中⁴。

(Conneau等人，2017b)。在WMT-14的法英语翻译测试集上，GPT-2能够凭借其强大的英文语言模型能力表现出色，取得了11.5分的BLEU分数。这一成绩优于（Artetxe等人，2017）以及（Lample等人，2017）提出的几种无监督机器翻译基准方法，但仍远低于目前最好的无监督机器翻译方案（Artetxe等人，2019）所取得的33.5分BLEU分数。由于我们在处理WebText数据时刻意去除了其中的非英文网页作为过滤步骤，因此该模型在该任务上的表现令我们颇为意外。为验证这一点，我们在WebText上运行了字节级的语言检测工具，结果仅检测到10MB的法语数据，这一数量大约是以往无监督机器翻译研究中常用的单语法语语料库大小的1/500。

3.8. 问答任务

检验语言模型内部包含哪些信息的一种可行方法，是评估它针对事实类问题生成正确答案的频率。此前在所有信息都存储于参数之类的神经网络系统中，已有研究展示了这种行为一种神经网络对话模型（Vinyals & Le, 2015）由于缺乏高质量的评价数据集，仅能给出定性结果。而近期推出的自然问题数据集（Kwiatkowski等人，

⁴<https://github.com/CLD2Owners/cld2>

2019) 是一种极具潜力的资源，可用于更定量地测试相关性能。与翻译任务类似，语言模型的上下文会通过示例问答对进行初始化，这有助于模型掌握该数据集要求的简短答案风格。根据在SQUAD等阅读理解数据集上常用的精确匹配度量标准进行评估时，GPT-2仅能正确回答4.1%的题目。³ 作为对比，最小的模型其准确率也仅能达到1.0%，这一数值对应的是一种极为简单的基线模型——它会针对每种问题类型（谁、什么、哪里等）直接返回出现频率最高的答案。由此可见GPT-2的正确作答率是该基线模型的5.3倍，这说明到目前为止，模型容量仍是制约神经网络在这类任务上表现不佳的核心因素。GPT-2为其生成的答案所赋予的概率值具有很好的校准性，且在它最为有信心那1%的题目中，其准确率可达63.1%。GPT-2在开发集问题上生成的30个最可信的答案展示在表5中。即便如此，GPT-2的性能依然远远落后于那些将信息检索与抽取式文档问答技术相结合的开放领域问答系统，后者的准确率通常在30%到50%之间Alberti等人，2019。

³原本自认为擅长随机知识问答的亚历克，在与GPT-2相同的测试环境下，对100个随机选取的示例进行了回答，其中正确答案有17个。实际上他只答对了14道题，那另外3道本应是他能做对的。

Language Models are Unsupervised Multitask Learners						
	PTB	WikiText-2	enwik8	text8	Wikitext-103	1BW
Dataset train	2.67%	0.66%	7.50%	2.34%	9.09%	13.19%
WebText train	0.88%	1.63%	6.31%	3.94%	2.42%	3.75%

Table 6. Percentage of test set 8 grams overlapping with training sets.

4. Generalization vs Memorization

Recent work in computer vision has shown that common image datasets contain a non-trivial amount of near-duplicate images. For instance CIFAR-10 has 3.3% overlap between train and test images (Barz & Denzler, 2019). This results in an over-reporting of the generalization performance of machine learning systems. As the size of datasets increases this issue becomes increasingly likely which suggests a similar phenomena could be happening with WebText. Therefore it is important to analyze how much test data also shows up in the training data.

To study this we created Bloom filters containing 8-grams of WebText training set tokens. To improve recall, strings were normalized to contain only lower-cased alphanumeric words with a single space as a delimiter. The Bloom filters were constructed such that the false positive rate is upper bounded by $\frac{1}{10^8}$. We further verified the low false positive rate by generating 1M strings, of which zero were found by the filter.

These Bloom filters let us calculate, given a dataset, the percentage of 8-grams from that dataset that are also found in the WebText training set. Table 6 shows this overlap analysis for the test sets of common LM benchmarks. Common LM datasets’ test sets have between 1-6% overlap with WebText train, with an average of overlap of 3.2%. Somewhat surprisingly, many datasets have larger overlaps with their own training splits, with an average of 5.9% overlap.

Our approach optimizes for recall, and while manual inspection of the overlaps shows many common phrases, there are many longer matches that are due to duplicated data. This is not unique to WebText. For instance, we discovered that the test set of WikiText-103 has an article which is also in the training dataset. Since there are only 60 articles in the test set there is at least an overlap of 1.6%.⁴ Potentially more worryingly, 1BW has an overlap of nearly 13.2% with its own training set according to our procedure.

For the Winograd Schema Challenge, we found only 10 schemata which had any 8-gram overlaps with the WebText training set. Of these, 2 were spurious matches. Of the remaining 8, only 1 schema appeared in any contexts that

⁴A significant portion of additional overlap is due to editors reusing some paragraphs across multiple articles with a shared theme such as various battles in the Korean War.

gave away the answer.

For CoQA, about 15% of documents in the news domain are already in WebText and the model performs about 3 F1 better on these. CoQA’s development set metric reports the average performance over 5 different domains and we measure a gain of about 0.5-1.0 F1 due to overlap across the various domains. However, no actual training questions or answers are in WebText since CoQA was released after the cutoff date for links in WebText.

On LAMBADA, the average overlap is 1.2%. GPT-2 performs about 2 perplexity better on examples with greater than 15% overlap. Recalculating metrics when excluding all examples with any overlap shifts results from 8.6 to 8.7 perplexity and reduces accuracy from 63.2% to 62.9%. This very small change in overall results is likely due to only 1 in 200 examples having significant overlap.

Overall, our analysis suggests that data overlap between WebText training data and specific evaluation datasets provides a small but consistent benefit to reported results. However, for most datasets we do not notice significantly larger overlaps than those already existing between standard training and test sets, as Table 6 highlights.

Understanding and quantifying how highly similar text impacts performance is an important research question. Better de-duplication techniques such as scalable fuzzy matching could also help better answer these questions. For now, we recommend the use of n-gram overlap based de-duplication as an important verification step and sanity check during the creation of training and test splits for new NLP datasets.

Another potential way of determining whether the performance of WebText LMs is attributable to memorization is inspecting their performance on their own held-out set. As shown in Figure 4, performance on both the training and test sets of WebText are similar and improve together as model size is increased. This suggests even GPT-2 is still underfitting on WebText in many ways.

GPT-2 is also able to write news articles about the discovery of talking unicorns. An example is provided in Table 13.

5. Related Work

A significant portion of this work measured the performance of larger language models trained on larger datasets. This

语言模型属于无监督多任务学习器。						
	PTB	WikiText-2数据集	enwik8数据集	text8数据集	Wikitext-103测试集	1BW
训练数据集	2.67%	0.66%	7.50%	2.34%	9.09%	13.19%
WebText训练集	0.88%	1.63%	6.31%	3.94%	2.42%	3.75%

表6. 测试集的8-gram测试数据与各训练集之间的重叠比例。

4. 泛化能力与记忆效应

计算机视觉领域的最新研究显示，常见的图像数据集中存在相当数量的近乎重复的图像。例如，CIFAR-10数据集的训练图像与测试图像之间的重叠比例就高达3.3%（Barz与Denzler, 2019）。这种情况会导致机器学习系统的泛化能力被高估。随着数据集规模的不断扩大，这类问题出现的概率也会随之上升，因此WebText数据集很可能也存在类似的情况。正因如此，分析训练数据中包含多少测试数据就显得尤为重要。

为研究这一问题，我们构建了包含WebText训练集标记对应的8-gram序列的布隆过滤器。为提升召回率，这些字符串被规范化处理，仅保留小写的字母数字字符，并以单个空格作为分隔符。这类布隆过滤器的假阳性率被限制在 $\frac{1}{10^8}$ 以内。此外，我们还通过生成100万个字符串进行了验证，结果发现该过滤器并未识别出任何错误匹配项，进一步证明了其较低的假阳性率。

借助这些布隆过滤器，我们可以计算给定数据集中有哪些8-gram序列也存在于WebText训练集中。表6展示了常见语言模型基准测试中各测试集的这种重叠分析结果。常见的各类语言模型数据集的测试集与WebText训练集之间的重叠比例在1%到6%之间，平均重叠比例为3.2%。令人略感意外的是，许多数据集与其自身的训练子集之间的重叠比例反而更高，平均重叠比例为5.9%。

我们的方法以召回率为优化目标，虽然对重叠内容进行人工检查后会发现许多常见短语，但也存在大量因数据重复而产生的较长匹配项。这种情况并非WebText数据集所独有。例如，我们发现WikiText-103的测试集中有一篇文章也出现在训练数据集中。由于测试集仅包含60篇文章，因此至少存在1.6%的重叠率。⁴ 更令人担忧的是，根据我们的检测方法，1BW数据集与其自身的训练数据集的重叠率竟接近13.2%。

在Winograd模式挑战赛中，我们仅发现10种模式与WebText训练集存在8-gram序列重叠度。其中2种属于虚假匹配，而在剩余的8种模式中，只有一种模式出现在某些上下文中。

⁴额外的重叠现象在很大程度上是由于编辑们在多篇主题相近的文章中重复使用了某些段落，比如关于朝鲜战争中各类战役的内容。

给出了答案。

对于CoQA数据集而言，新闻领域中约15%的文档已经包含在WebText数据集中，模型在这些文档上的F1分数能提升大约3分。CoQA的测试集指标是依据5个不同领域的平均表现来评估的，由于跨领域之间的重叠效应，我们的模型F1分数能够提升约0.5到1.0分。不过需要注意的是，由于CoQA的发布时间晚于WebText链接收录的截止日期，因此WebText中并不包含任何实际的训练题或答案。

在LAMBADA数据集上，平均重叠率为1.2%。而对于重叠率超过15%的示例，GPT-2的困惑度表现要好约2点。如果在排除所有存在重叠的示例后重新计算指标，其困惑度将从8.6上升至8.7，准确率则从63.2%下降到62.9%。整体结果的这种微小变化很可能是由于仅有1/200的示例存在显著重叠所致。

总体而言，我们的分析表明，WebText训练数据与特定评估数据集之间的数据重叠虽然幅度不大，但能够为所得结果带来持续的正向影响。不过，如6中所显示的，对于大多数数据集而言，其数据重叠程度并没有显著超过常规训练集与测试集之间的重叠水平。

了解并量化文本相似度对性能的影响程度是一个重要的研究课题。诸如可扩展的模糊匹配这类更高效的去重处理技术，也有助于更好地回答这些问题。目前，我们建议在为新NLP数据集划分训练集和测试集时，采用基于n-gram序列重叠度的去重处理方法，以此作为重要的验证步骤和合理性检查手段。

另一种判断WebText语言模型的性能是否源于记忆效应的方法，是考察其在预留的独立测试集上的表现。如4图所示，WebText语言模型在训练集和测试集上的性能较为接近，且会随着模型规模的增大而一同提升。这一现象表明，即便是在很多方面已经相当强大的GPT-2，在处理WebText数据时依然存在欠拟合的问题。

GPT-2同样能够撰写关于会说话的独角兽被发现的新闻报道，表中给出了相关示例13。

5. 相关工作

本研究中有相当一部分内容用于评估在更大数据集上训练的更大规模语言模型的性能。

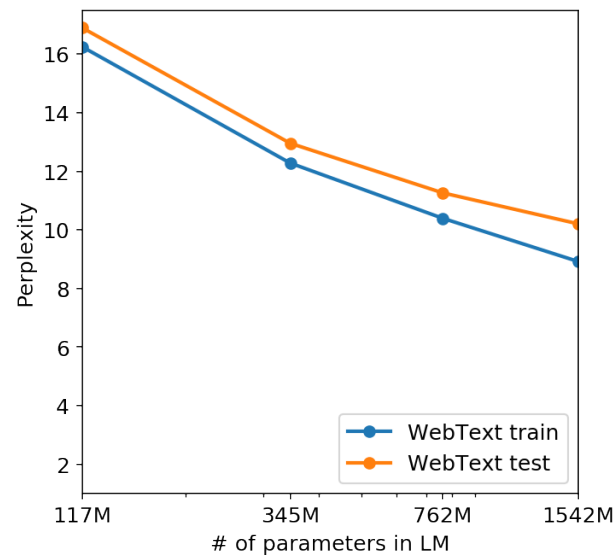


Figure 4. The performance of LMs trained on WebText as a function of model size.

is similar to the work of Jozefowicz et al. (2016) which scaled RNN based language models on the 1 Billion Word Benchmark. Bajgar et al. (2016) also previously improved results on the Children’s Book Test by creating a much larger training dataset out of Project Gutenberg to supplement the standard training dataset. Hestness et al. (2017) conducted a thorough analysis of how the performance of various deep learning models changes as a function of both model capacity and dataset size. Our experiments, while much noisier across tasks, suggest similar trends hold for sub-tasks of an objective and continue into the 1B+ parameter regime.

Interesting learned functionality in generative models has been documented before such as the cells in an RNN language model performing line-width tracking and quote/comment detection Karpathy et al. (2015). More inspirational to our work was the observation of Liu et al. (2018) that a model trained to generate Wikipedia articles also learned to translate names between languages.

Previous work has explored alternative approaches to filtering and constructing a large text corpus of web pages, such as the iWeb Corpus (Davies, 2018).

There has been extensive work on pre-training methods for language tasks. In addition to those mentioned in the introduction, GloVe (Pennington et al., 2014) scaled word vector representation learning to all of Common Crawl. An influential early work on deep representation learning for text was *Skip-thought Vectors* (Kiros et al., 2015). McCann et al. (2017) explored the use of representations derived from machine translation models and Howard & Ruder (2018)

improved the RNN based fine-tuning approaches of (Dai & Le, 2015). (Conneau et al., 2017a) studied the transfer performance of representations learned by natural language inference models and (Subramanian et al., 2018) explored large-scale multitask training.

(Ramachandran et al., 2016) demonstrated that seq2seq models benefit from being initialized with pre-trained language models as encoders and decoders. More recent work has shown that LM pre-training is helpful when fine-tuned for difficult generation tasks like chit-chat dialog and dialog based question answering systems as well (Wolf et al., 2019) (Dinan et al., 2018).

6. Discussion

Much research has been dedicated to learning (Hill et al., 2016), understanding (Levy & Goldberg, 2014), and critically evaluating (Wieting & Kiela, 2019) the representations of both supervised and unsupervised pre-training methods. Our results suggest that unsupervised task learning is an additional promising area of research to explore. These findings potentially help explain the widespread success of pre-training techniques for down-stream NLP tasks as we show that, in the limit, one of these pre-training techniques begins to learn to perform tasks directly without the need for supervised adaption or modification.

On reading comprehension the performance of GPT-2 is competitive with supervised baselines in a zero-shot setting. However, on other tasks such as summarization, while it is qualitatively performing the task, its performance is still only rudimentary according to quantitative metrics. While suggestive as a research result, in terms of practical applications, the zero-shot performance of GPT-2 is still far from use-able.

We have studied the zero-shot performance of WebText LMs on many canonical NLP tasks, but there are many additional tasks that could be evaluated. There are undoubtedly many practical tasks where the performance of GPT-2 is still no better than random. Even on common tasks that we evaluated on, such as question answering and translation, language models only begin to outperform trivial baselines when they have sufficient capacity.

While zero-shot performance establishes a baseline of the potential performance of GPT-2 on many tasks, it is not clear where the ceiling is with finetuning. On some tasks, GPT-2’s fully abstractive output is a significant departure from the extractive pointer network (Vinyals et al., 2015) based outputs which are currently state of the art on many question answering and reading comprehension datasets. Given the prior success of fine-tuning GPT, we plan to investigate fine-tuning on benchmarks such as decaNLP and GLUE, especially since it is unclear whether the additional

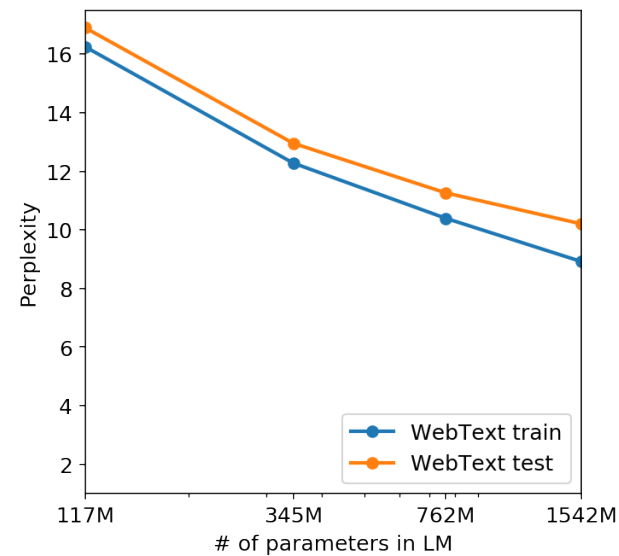


图4. 基于WebText数据集训练得到的语言模型的性能随模型规模的变化情况。

这与 Jozefowicz等人 (2016年) 的研究类似, 后者基于10亿词基准测试对基于RNN的语言模型进行了优化。Bajgar等人 (2016年) 也曾通过利用古腾堡计划中的大量资源构建出规模更大的训练数据集来补充常规训练数据集, 从而提升了在儿童书籍测试中的表现。Hestness等人 (2017年) 对不同深度学习模型的性能如何随模型容量与数据集规模的变化而变化进行了深入分析。尽管我们的实验在不同任务间的结果波动较大, 但仍显示出该目标下的各子任务也遵循类似的趋势, 并且这种趋势会延续到10亿参数量级。+的范围内。

此前已有研究记录过生成模型中出现的有趣学习特性, 例如RNN语言模型中的单元能够实现行宽跟踪以及引文/注释检测功能Karpathy等人(2015)。对我们这项工作启发更大的, 则是Liu等人(2018)发现的这样一个现象: 那些被训练用于生成维基百科文章的模型, 还能学会在不同语言之间进行名称翻译。

以往的研究还探索过构建大型网页文本语料库的其他方法, 比如iWeb语料库 (Davies, 2018)。

在针对语言任务的预训练方法方面, 已有大量研究开展。除了引言中提到的那些方法外, GloVe (Pennington等人, 2014年) 还将词向量表示的学习方法应用到了整个Common Crawl数据集上。早期在文本深度表示学习领域具有重大影响力的研究成果便是跳思向量模型 (Kiros等人, 2015年)。McCann等人 (2017年) 则探讨了利用从机器翻译模型中提取的表示方法, 而Howard与Ruder (2018年) 也进行了相关研究。

他们改进了 (Dai与Le, 2015年) 提出的基于RNN的微调方法。(Conneau等人, 2017a) 研究了通过自然语言推理模型学到的表示方法的迁移性能, 同时Subramanian等人 (2018年) 则探索了大规模多任务训练的可能性。

(Ramachandran等人, 2016)证明, 将预训练语言模型作为编码器与解码器来对Seq2Seq模型进行初始化, 能够提升其性能。更近期的研究也表明, 即便是在针对闲聊对话、基于对话的问答系统这类难度较高的生成任务进行微调时, 进行LM预训练也同样十分有效 (Wolf等人, 2019) (Dinan等人, 2018)。

6. 讨论

已有大量研究致力于 (Hill等人, 2016年) 对有监督和无监督预训练方法的呈现进行学习 (Levy与Goldberg, 2014年)、理解 (Wieting与Kiela, 2019年) 以及批判性评估。我们的研究结果表明, 无监督任务学习是另一个极具潜力的研究方向。这些发现有助于解释为何预训练技术在各类下游自然语言处理任务中能取得如此广泛的成功——因为我们在极限情况下发现, 其中某种预训练技术最终能够直接完成任务, 而无需经过有监督的调整或修改。

在阅读理解任务上, GPT-2在零样本场景下的表现能与有监督的基准模型相媲美。不过在摘要生成任务等其他任务中, 尽管其任务完成质量尚可, 但根据定量指标来看, 其表现依然十分初级。这一研究结果虽具参考价值, 但从实际应用角度来看, GPT-2在零样本场景下的表现仍远未达到可用水平。

我们已研究了WebText语言模型在众多典型自然语言处理任务上的零样本表现, 但实际上还有许多其他任务值得评估。毫无疑问, 在很多实际应用场景中, GPT-2的性能依然不比随机猜测更好。即便是在我们之前评估过的常见任务, 比如问答任务和翻译任务, 只有当语言模型具备足够的计算能力时, 才能开始超越那些极其简单的基准模型。

虽然零样本性能为GPT-2在诸多任务上的潜在表现设定了基准, 但通过微调其表现上限能达到何种程度仍不明确。在某些任务中, GPT-2生成的完全抽象化输出与基于提取式指针网络的输出存在显著差异 (Vinyals等人, 2015年), 而后者目前在许多问答任务和阅读理解数据集上均已达到当前最佳水平。鉴于此前对GPT进行微调所取得的成功, 我们计划在decaNLP和GLUE等基准测试上开展相关研究, 尤其是因为目前尚不清楚额外的微调是否

Language Models are Unsupervised Multitask Learners	
	
training data and capacity of GPT-2 is sufficient to overcome the inefficiencies of uni-directional representations demonstrated by BERT (Devlin et al., 2018).	
7. Conclusion	
When a large language model is trained on a sufficiently large and diverse dataset it is able to perform well across many domains and datasets. GPT-2 zero-shots to state of the art performance on 7 out of 8 tested language modeling datasets. The diversity of tasks the model is able to perform in a zero-shot setting suggests that high-capacity models trained to maximize the likelihood of a sufficiently varied text corpus begin to learn how to perform a surprising amount of tasks without the need for explicit supervision. ⁵	
Acknowledgements	
Thanks to everyone who wrote the text, shared the links, and upvoted the content in WebText. Many millions of people were involved in creating the data that GPT-2 was trained on. Also thanks to all the Googlers who helped us with training infrastructure, including Zak Stone, JS Riehl, Jonathan Hseu, Russell Power, Youlong Cheng, Noam Shazeer, Solomon Boulos, Michael Banfield, Aman Gupta, Daniel Sohn, and many more. Finally thanks to the people who gave feedback on drafts of the paper: Jacob Steinhardt, Sam Bowman, Geoffrey Irving, and Madison May.	
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^[5] Preliminary code for downloading and using the small model is available at https://github.com/openai/gpt-2

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7. 结论	
当大型语言模型在规模足够庞大且种类多样的数据集上经过训练后，它便能在众多领域和数据集中展现出优异的性能。GPT-2在经过测试的8个语言模型相关数据集中的7个上，均实现了接近当前最佳水平的零样本性能。该模型在零样本场景下能够处理的任务种类之丰富，表明那些为最大化处理多样化文本语料库的可能性而设计的高容量模型，已经开始学会在无需显式监督的情况下完成大量不同的任务。 ⁵	
致谢	
感谢所有为WebText撰写文章、分享链接以及为相关内容点赞的人士。还有数以百万计的人参与了构建GPT-2所依赖的训练数据。同时也要感谢所有在训练基础设施方面给予我们帮助的谷歌员工，其中包括Zak Stone、JS Riehl、Jonathan Hseu、Russell Power、Youlong Cheng、Noam Shazeer、Solomon Boulos、Michael Banfield、Aman Gupta、Daniel Sohn以及更多人。最后，感谢那些对论文初稿提出反馈意见的人：Jacob Steinhardt、Sam Bowman、Geoffrey Irving和Madison May。	
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语言模型属于无监督多任务学习器	
该研究探讨了将用多种特征集进行内容提取的方法。相关成果发表于第22届国际万维网大会论文集，第89–90页，美国计算机协会，2013年。作者为彼得斯, M. E.、诺伊曼, M.、伊耶尔, M.、加德纳, M.、克拉克, C.、李K以及泽特尔莫耶, L，论文题为《深度上下文化词表示》。相关预印本可见 arXiv 预印本 <i>arXiv:1802.05365</i> ，2018年。此外还有拉德福德, A.、约泽福维奇, R.和苏茨克弗, I所著的《生成评论与情感分析学习》，对应预印本为 arXiv 预印本 <i>arXiv:1704.01444</i> ，2017年；拉德福德, A.、纳拉辛汉, K.、萨利曼斯, T.和苏茨克弗, I撰写的《通过生成式预训练提升语言理解能力》，发表于2018年；拉马钱德兰, P.、刘鹏杰和黎启伟共同完成的《用于序列到序列学习的无监督预训练》对应预印本为 arXiv 预印本 <i>arXiv:1611.02683</i> ，2016年。还有雷赫特, B.、罗洛夫斯, R.、施密特, L.和尚卡尔, V发表的《CIFAR-10分类器能否泛化到CIFAR-10数据集？》，相关预印本见 arXiv 预印本 <i>arXiv:1806.00451</i> ，2018年；雷迪, S.、陈丹和曼宁, C. D提出的CoQA：一种对话式问答挑战，其预印本为 arXiv 预印本 <i>arXiv:1808 07042</i> ，2018年，文末还有.标记。施瓦茨, R.、萨普, M.、康斯塔斯, I.、齐勒斯, L.、崔Y和史密斯, N. A完成了故事填空任务的相关研究，相关成果收录于第二届词汇、句子及篇章层面语义模型关联研讨会论文集，第52–55页，2017年。西, A.、刘鹏杰和曼宁, C. D提出了基于指针生成网络的新闻摘要方法，对应预印本为 arXiv 预印本 <i>arXiv:1704.04368</i> ，2017年。森里奇, R.、哈多ウ, B.和伯奇, A研究了使用子词单元对罕见词进行神经机器翻译的技术，相关预印本为 arXiv 预印本 <i>arXiv:1508.07909</i> ，2015年。苏布拉马尼安, S.、特里施勒, A.、本吉奥, Y.和帕尔, C. J通过大规模多任务学习实现了通用分布式句子表示的学习，相关预印本为 arXiv 预印本 <i>arXiv:1804 00079</i> ，2018年，文末同样有.标记。苏茨克弗, I.、维尼亚尔斯, O.和黎启伟发表了《基于神经网络的序列到序列学习》一文，该文收录于神经信息处理系统进展会议，第3104–3112页，2014年。此外他们还共同撰写了《迈向原理性的无监督学习》，对应预印本为 arXiv 预印本 <i>arXiv:1511.06440</i> ，2015年。特里切尔艾尔, P.、埃马米, A.、张建昌、特里施勒, A.、苏莱曼, K.和迪亚兹, F研究了自然语言理解中常识推理的评估方法，相关预印本为 arXiv 预印本 <i>arXiv:1811.01778</i> ，2018年。郑廷辉和黎启伟提出了一种简单的常识推理方法，相关预印本为 arXiv 预印本 <i>arXiv:1806.02847</i> ，2018年。瓦斯瓦尼, A.、沙齐尔, N.、帕尔马尔, N.、乌什科雷特, J.、琼斯, L.、戈麦斯, A. N.、Ł.凯泽和波洛苏欣, I发表了《注意力就是你所需》一文，该文收录于神经信息处理系统进展会议，第5998–6008页，2017年。最后维尼亚尔斯, O.和黎启安共同提出了一个神经网络对话模型，相关预印本为 arXiv 预印本 <i>arXiv:1506.05869</i> ，2015 年。	O·维尼亚尔斯、马可·福图纳托与尼古拉斯·贾特利共同发表了《指针网络》一文。该文收录于神经信息处理系统进展会议，刊载于2015年的相关论文集中，页码为2692–2700页。
	王A.、辛格A.、迈克IJ.、希尔F.、奥利弗·莱维以及S. R.鲍曼共同提出了GLUE项目——这是一个用于自然语言理解的多任务基准测试与分析平台。 arXiv 预印本平台 <i>arXiv:1804.07461</i> 上的相关内容发表于2018年。
J·韦斯顿《神经信息处理的建造者会议》中，相关内容刊载于2016年，页码为829–837页。	
J·维廷与D·基拉在2019年发表了论文《无需训练：探索用于句子分类的随机编码器》，相关预印本发布于arXiv预印本平台，编号为arXiv:1901.10444。同年，T·沃尔夫、V·桑赫、J·肖蒙德以及C·德拉昂格共同提出了基于迁移学习的神经网络对话Agent方法，该研究对应的预印本编号为arXiv:1901.08149。此外，D·约加塔玛、C·D·M·达图姆、J·康纳、T·科西斯基、M·赫尔扎诺夫斯基、L·孔、A·拉扎里杜、W·凌、L·余、C·戴尔等学者也在2019年发表了探讨通用语言智能学习与评估方法的论文，其预印本编号为arXiv:1901.11373。	

8. Appendix A: Samples

8.1. Model capacity

To complement the reported perplexity gains of bigger LMs on WebText show in Figure 4, Tables 7 through 11 show side-by-side completions of the smallest WebText LM and GPT-2 on random unseen WebText test set articles.

8.2. Text Memorization

We observe some memorizing behavior in GPT-2 on longer strings that are repeated many times in the dataset such as famous quotes or speeches. For example, when conditioned on the first sentence and a half of the Gettysburg Address (which occurs approximately 40 times throughout WebText), an argmax decode from GPT-2 recovers the speech. Even when sampling without truncation, we find that the model copies the speech for awhile before drifting, albeit in a similar style. It typically drifts within 100-200 tokens, and displays widening diversity once it drifts.

To quantify how often exact memorization shows up in samples, we generated samples from GPT-2 conditioned on WebText test set articles and compared the overlap rates of GPT-2’s generations to the overlap rates of the ground-truth completions. The results of this analysis are shown below and suggest that GPT-2 repeats text from the training set less often then the baseline rate of held-out articles.

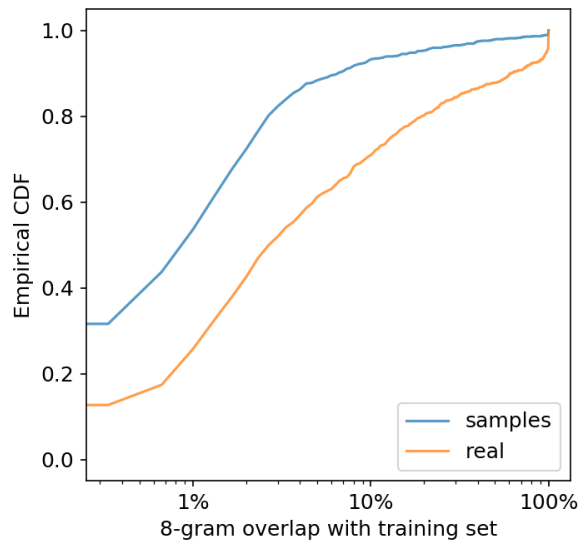


Figure 5. CDF of percentage 8-gram overlap with WebText training set, for both WebText test set and samples (conditioned on WebText test set, with top- k truncated random sampling with $k = 40$). Most samples have less than 1% overlap, including over 30% of samples with no overlap, whereas the median for test set is 2.6% overlap.

8.3. Diversity

Table 12 shows multiple completions of the same random WebText test set context, showing the diversity of completions with standard sampling settings.

8.4. Robustness

Table 13 shows the previously mentioned talking unicorns news article. We find the model to be capable of handling out of distribution contexts, but the quality of these samples is generally lower.

8. 附录A：样本

8.1. 模型容量

为补充图中所示更大规模语言模型在WebText数据集上所取得的困惑度下降成果4，相关表格7至11展示了最小规模WebText语言模型与GPT-2在随机未见过的语料集文章上的补全结果对比。

8.2. 文本记忆现象

在数据集中出现多次重复的较长字符串——诸如名人名言或演讲内容时，我们观察到GPT-2会展现出一定的记忆行为。例如，当以《葛底斯堡演说》的前半段作为条件输入时（该段落落在WebText数据集中大约出现了40次），通过argmax解码方式就能还原出完整的演说内容。即便在不进行截断处理的采样过程中，该模型也会在一段时间内复制这段演说内容，随后虽然风格仍会保持相似，但内容会逐渐偏离。通常情况下，这种偏离会在100到200个标记之内发生，且一旦开始偏离，其生成的文本多样性也会随之增加。

为量化样本中出现纯粹记忆效应的频率，我们基于WebText测试集的文章作为条件，从GPT-2中生成了相应样本，并将GPT-2生成的文本与真实正确结果的重叠率进行对比。该分析的结果如下所示，表明GPT-2重复使用训练集中的文本的频率，低于那些未被用于训练的样本的基准频率。

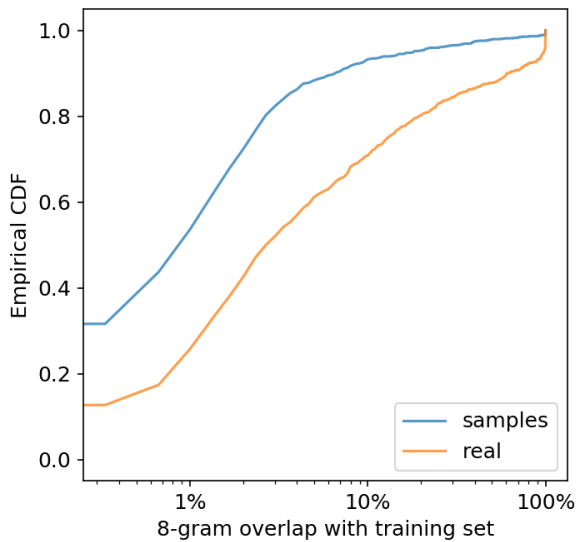


图5. WebText训练集以及测试集和样本的百分比8-gram重叠度的累积分布函数（条件为WebText测试集，采用顶部 k 个标记截断后的随机采样方法，采样比例为 $k = 40$ ）。大多数样本的重叠度低于1%，其中超过30%的样本完全没有重叠，而测试集的中间值则为2.6%的重叠度。

8.3. 多样性

表格12展示了针对同一组随机选取的WebText测试集上下文所生成的多种补全结果，体现了在标准采样设置下这些补全结果的多样性。

8.4. 鲁棒性

表格13展示了前面提到的“会说话的独角兽”新闻文章。研究显示该模型能够处理处于上下文之外的内容，但这些样本的质量通常较低。

Context (WebText test)	
Outfit 1: Typical	
This pairing was the first outfit I thought of when I bought the shoes. It’s like a summer version of this Jake Grantham outfit; in fact, my shoes are close to the colors of his Nike Racers! Instead of a heavy Harris Tweed jacket and denim shirt, I’m wearing a cotton DB jacket and and a linen shirt. Both fabrics (in these colors) are an absolute must for summer, as they go with both dark and and light pants! As you can see, they pair wonderfully with the dark jeans and shoes. It’s a pseudo menswear/prep outfit.	
Overall, this is a very casual outfit which is why I paired my sneakers with it. I’m not about wearing a full wool suit with sneakers (as GQ shows a lot) but I’m definitely open to keeping things casual, like this cotton DB. Casual fabrics are key to pulling off your sneakers in a dressed down menswear outfit. I’d even suggest to wear these sneakers with a khaki chino suit or a white linen suit. Just be sure to ditch the tie or wear a tee or polo; wearing a tie with sneakers is a bit too much	
Small model completion	GPT-2 completion
.	for me!
Outfit 2: High End	Outfit 2: Casual Jacket
This outfit was just in the eye of the storm. It’s a hybrid of what ASICS does with streetwear (like J. Crew) and what J. Crew does with casual pieces that are a bit more traditional. Again, as I mentioned before, it’s a low budget item and they can be very expensive. These will definitely be an issue if you’re trying to work as a casual sneakermaker, and that’s OK but don’t hesitate if you want to try the lower price versions of their outfits!	I know that some people might be opposed to wearing sneakers in a turtleneck, but I wanted to be true to myself, so I went with a slim fitting turtleneck in a color more similar to my favorite color of the day. Overall, it’s a classic turtleneck, with nothing too flashy. I did swap my shoes for these ASICS Tiger Killshots and I wish I hadn’t did that. The shoe itself is very comfortable, however, I found that wearing them in these sneakers made my feet look a bit larger. I’d wear them with denim, chinos, whatever, but would never recommend wearing them alone. There are just too many variables involved.
	Outfit 3: Dress
	This is the final outfit I wore the ASICS Tiger Killshots. This version of the ASICS is a much more modern and streamlined version. The shoe has the most modern and minimal design that I can think of! With a lot of black, these sneakers look very cool. They’ll definitely be very comfortable for summer.
Outfit 3: High Light	
This was a ”high light” look while I had a chance to get some color. I think the color is based on	To me, wearing them with a suit is the easiest way to have a good pair of sneaker shoes. That’s pretty much it! If

Table 7. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top- k random sampling with $k = 40$ was used for generation.

上下文（WebText测试）	
服装1：常规款	
购买这双鞋时，我首先想到的搭配就是这服饰。它可以说是Jake Grantham那套穿搭的夏季版本；事实上，我的鞋子颜色和他的Nike Racers系列极为相近！我没有选择厚重的哈里斯粗花呢夹克和牛仔衬衫，而是穿了一件棉质DB夹克搭配亚麻衬衫。这类面料（以这样的颜色呈现）绝对是夏季的必备之选，因为它们既适合深色裤子，也适合浅色裤子！如您所见，它们与深色牛仔裤和鞋子搭配起来非常出色。这其实是一种类男士正装/预备装风格的穿搭。	
总体而言，这是一套非常休闲的穿搭，这也是我将运动鞋与之搭配的原因。我不喜欢像《GQ》杂志经常展示的那样，穿全羊毛西装搭配运动鞋，但我当然乐于保持这种休闲风格，比如这套棉质DB系列服饰。休闲面料是让运动鞋在轻松的男士正装造型中显得自然得体的关键。我甚至建议可以将这双运动鞋与卡其色Chino西装或白色亚麻西装搭配穿着。只要记得不要系领带，或者改穿T恤或马球衫即可；毕竟运动鞋搭配领带会显得有点过于正式。	
小型模型补全功能	GPT-2文本生成功能
.	是专为我设计的！
服装系列2：高端款	穿搭2：休闲夹克
这套服装恰好处于风暴的中心地带。它是一种ASICS在街头服饰领域的设计风格（类似J. Crew）与J. Crew在休闲单品方面的设计风格的结合体。以及J. Crew在那些略带独特风格的休闲单品上的设计思路的融合。更为传统的方式。再说一次，正如我之前所提到的，这种方式的成本较低。这类项目属于预算支出项，且往往价格十分昂贵。如果你打算以临时工的身份工作，这无疑会成为一个问题。从事运动鞋设计工作也没关系，但如果你有其他选择的话，不妨考虑一下。如果你想尝试他们价格更低的服装系列，那也很不错！	我知道可能有人反对在穿高领衫时搭配运动鞋。不过我希望能忠于自己的风格。因此我选择了一件剪裁合身的高领衫，颜色也与我当天最喜欢的颜色相近。整体而言，这便是如此。就像我今日最喜欢的颜色一样。没错，就是一件普通的经典高领衫，没有特别引人注目的地方。我确实更换了我的……给这些ASICS Tiger Killshots穿的鞋，真希望自己当初没买。我确实这么做了。不过，这双鞋本身的舒适度很高。我发现穿在运动鞋里面会让我的脚看起来稍大一些。我会搭配牛仔裤或卡其裤来穿它们，不过无论怎么搭配，都不会推荐单独穿着。毕竟其中涉及的因素实在太多，风险太高了。单独穿的话问题会更多。
	服装3：连衣裙
	这就是我当时穿的最终款服装。ASICS Tiger Killshots系列。这款ASICS产品是更为现代且设计更为简洁的版本。它的设计极为现代化且极简，是我所能想到的最出色的款式之一。它拥有我所能想象到的最为现代且极简的设计风格。这双黑色运动鞋看起来非常酷炫，肯定很棒。在夏天穿的话，也会十分舒适。
穿搭3：亮点造型	
这是我有机会尝试一些亮色时打造的“亮点”造型。我觉得这款颜色的设计灵感来源于……	对我来说，将它们与西装搭配是拥有一双出色运动鞋的最简单方式。差不多就是这样了——只要这样搭配，就能拥有一双很棒的运动鞋。就是它了！如果……

表7. 随机选取的未见上下文（上方），以及来自最小模型（左侧）和最大模型（右侧）的非筛选补全结果。这些上下文的总标记数为768，其中约256个标记对应的段落会被展示；而补全结果的长度为256个标记，会完整显示出来。生成过程中采用了基于 k 随机排序的 $k = 40$ 采样方法。

Context (WebText test) Some of the most glorious historical attractions in Spain date from the period of Muslim rule, including The Mezquita, built as the Great Mosque of Córdoba and the Medina Azahara, also in Córdoba and now in ruins but still visitable as such and built as the Madinat al-Zahra, the Palace of al-Andalus; and the Alhambra in Granada, a splendid, intact palace. There are also two synagogues still standing that were built during the era of Muslim Spain: Santa María la Blanca in Toledo and the Synagogue of Córdoba, in the Old City.	
Reconquista and Imperial era [edit]	
Small model completion	GPT-2 completion
The era of Muslim Spain is very different from that of other parts of the world, with a rather extensive medieval history. While the Spanish Empire existed during the second half of the 9th century BC it was quite slowly transformed to a small, secularized empire by 1150 A.D., in a much more prosperous and diverse state. In 1050 CE the emperor of Naples, at least partially in characterised by a series of wars and invasions from the north-western coast to the western coast of the region (e.g., on the Danube River which extended all the way from Gibraltar into the south of Spain, but was quickly destroyed on the same day by the Roman army), assumed the title of Emperor of Naples. He made a large contribution over the last few decades to restore and maintain the empire, the first of which to the west. His successors were Ferdinand, who took power from the Papal Throne in 1148, while the first imperial successor was King Alfred II of Sweden. The empire was again made one of the few independent states in the European Continent and although no more than 200 million people lived on its territory during this period the people had much to say to the Emperor. From 1720 on the Muslims made up half the population of Spain.	The Reconquista (literally, "The return to the country") of 1092 and 1095 saw the establishment in the Iberian peninsula of a united kingdom of Castile and Aragon in which a Muslim dynasty would exist. This was the last time in Europe where Muslim rule would be considered "native" and the last occasion where Christian-inflamed and organized conflicts would make themselves known in Europe. The reign of Muslim Moors in Spain is often referred to, especially among non-Muslims, as the "Conquest of Granada" and Spain is sometimes referred to as Spain in "Granada". The last years of Muslim rule were followed by the Spanish Reconquista where the Muslims were driven or compelled out of the Iberian Peninsula with the victory of Ferdinand IX. in the Battle of Poitiers in March 1189 and the conquest of Jerusalem in 1187 by Saladin. The Reconquista period in Spain came to an end when the Moors retreated north. However the Moorish rule would not be fully replaced by the Christian rule until the Reconquista era resumed when the Reconquista lasted from 1229 to 1492. The last Muslim state in Spain, Granada, was conquered by the

Table 8. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top-*k* random sampling with *k* = 40 was used for generation.

上下文 (WebText测试) 西班牙一些最令人赞叹的历史遗迹可追溯至穆斯林统治时期， 其中包括…… 科尔多瓦的大清真寺与麦地那·阿扎哈拉，二者均建于科尔多瓦，如今虽已沦为废墟，依然可以参观；此外还有安达卢斯宫殿——麦地那特·阿尔-扎赫拉， 以及位于格拉纳达的阿尔罕布拉宫，那是一座保存完好的宏伟宫殿。此外， 在穆斯林统治西班牙的时代还建有两座至今仍屹立的犹太会堂。托莱多的圣玛利亚拉布兰卡教堂以及老城区的科尔多瓦犹太会堂。	
收复失地运动与帝国时代 [编辑]	
小型模型补全	GPT-2文本生成功能
穆斯林统治下的西班牙与其它地区的情况大相径庭，那里拥有相当悠久的历史。 其他地区则拥有较为丰富的中世纪历史。 在西班牙帝国存在的时期， 即公元前9世纪后半叶， 该帝国的变革进程极为缓慢。直到公元1150年左右， 它才逐渐转变为一个规模较小且世俗化的帝国。此时它已发展成为一个更为繁荣且多元化的国家。公元1050年时，那不勒斯的国王， 其统治特征至少在一定程度上表现为来自西北海岸的一系列战争与入侵。该地区的西海岸（例如多瑙河沿岸）这条河流从直布罗陀一直延伸至该地区位于西班牙南部，但很快就在同一地点被摧毁在罗马军队于当日击败他之后，他便加冕为皇帝那不勒斯。在过去的几十年里，他为该城的发展作出了巨大贡献他为恢复与维护帝国付出了数十年的努力，成为首位如此做的君主其西部为……。他的继任者是费迪南德，他……于1148年从教皇宝座手中夺取了权力，而首位……该帝国的继任者是瑞典的阿尔弗雷德二世国王。……这个帝国再度成为欧洲大陆为数不多的独立国家之一。尽管在那个时期其境内居住的人口不超过200……在这一时期，有数百万人生活在其领土上。人们有许多话要对皇帝说。从1720年起，穆斯林占西班牙人口的一半。	“收复失地运动”（字面意思是“重返故土”）1092年和1095年间，伊比利亚半岛上建立了卡斯蒂利亚与阿拉贡的统一王国。在伊比利亚半岛上形成了卡斯蒂利亚与阿拉贡的统一王国。届时将存在一个穆斯林王朝，这也是最后的穆斯林统治时期。这是欧洲境内最后一段可被视为穆斯林统治的时光。这也是“本土”穆斯林统治的最后一次出现，同时也是基督教势力引发的有组织冲突首次显现的时期。 此外，有组织冲突也开始在这一时期在欧洲出现。——欧洲。 穆斯林王朝的统治时代。西班牙的摩尔人常被稱為所称的，尤其是在非穆斯林群体中，他们更常被如此称呼。 作为“格拉纳达征服战”与西班牙有时也会被提及在“格拉纳达”被称作西班牙。穆斯林统治的最后岁月其后便是西班牙的收复失地运动。穆斯林被迫离开伊比利亚半岛。西班牙的收复失地运动就此宣告结束。1189年3月的普瓦捷之战以及对耶路撒冷的征服1187年，被萨拉丁征服。 西班牙的收复失地运动时期就此结束了。当摩尔人向北方撤退时。然而，摩尔人的统治并未就此彻底被基督教统治所取代。直到……，基督教统治才完全取代原有的摩尔人统治。随着收复失地运动的重新开启，这一运动又持续了下去。时间跨度为1229年至1492年。 西班牙境内的最后一个穆斯林政权是…… 格拉纳达，它最终被……由……查询

表8。随机未见上下文（上方），以及来自最小规模模型（左侧）和最大规模模型（右侧）的未经筛选的补全结果。这些上下文的总标记数为768， 其中仅展示约256个标记对应的段落；而补全结果的长度为256个标记， 会完整显示出来。生成过程采用了基于的顶级随机采样策略，并结合进行操作。

Language Models are Unsupervised Multitask Learners	
Context (WebText test)	
Pinky	
The pink ghost’s AI is designed to ”feel” opposite of the red ghost’s behavior. Pinky actually attempts to get out in front of Pac-Man. This is accomplished by setting the target 4 tiles ahead of Pac-Man’s current location in the direction that Pac-Man is travelling. One exception to this is when Pac-Man is traveling up. Due to an overflow bug in the code, the calculation includes a left offset equal to the expected up offset.	
The use of Pac-Man’s expected position allows the player to exploit the navigation code for Pinky. This was most often seen in situations where Pac-Man and Pinky are in close proximity to each other. For example, if they were headed toward one another, once they are less than 4 tiles apart, the target tile is now behind Pinky. If he encounters an intersection while in this configuration, he will attempt to double-back (by turning) in order to reach the target tile.	
Player’s would often perform what was referred to as a ”head fake” by changing directions rapidly just as Pinky was approaching an intersection. By displacing the target tile just as Pinky	
Small model completion was coming to the left, they would get the ”correct” direction. If the location is actually farther from what Pac-Man was at, the player’s will not respond to the ghost’s movement, and the ghost will instead return to the player’s current location. This is because Pac-Man does not move by turning. The ghost will have to go to another location of some kind to return where he began, and this is achieved in many ways. It will often simply leave the location, as for example if Pac-Man did jump off a cliff, he would return to the spot where he was before vanishing into space. There is a good number of possibilities these ghost paths may be possible, but the most obvious ones are if the player wants to avoid getting caught up in the game, with a goal as easy as keeping Pac-Man in the game for 1/3 as long as he was. Chasing In Pac-Man’s case, this would be similar to how Pac-Man can ”catch up” in a fight. A ghost could chase through enemies. This is useful for many reasons, both for Pac-Man and his AI. Unlike most Ghost games, this ghost simply travels in the direction from	GPT-2 completion was about to cross the intersection, Pac-Man would be able to dodge the ghosts’ projectiles and return to the safe location of the safe house. Buster The yellow ghost’s AI is specifically designed to try to avoid Pac-Man from leaving the safe house. The player must press the SPACE bar to reveal the safe house’s doors, then go into them to find the yellow ghost hiding in the next room. The player must also note that the ghost will not attempt to exit the safe house if it sees Pac-Man already there. In fact, the ghost will not attempt an exit in the first place unless they’re already in the safe house. These two behaviors are based on the fact that the player does not always attempt to reach the safe house before Pac-Man (or Blinky). The player must also beware of this glitch if a ghost (or another player) is already in the safe house and attempting to get out as soon as Pac-Man or Blinky is gone. At this point, the ghost will attempt to exit the safe house in a similar manner as it is in ”Chase” mode. Pac-Man There are some ghosts that follow Pac-Man. These ghosts are

Table 9. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top-*k* random sampling with *k* = 40 was used for generation.

语言模型属于无监督多任务学习器。	
上下文（WebText测试）	
平基	
粉色幽灵所对应的人工智能被设计为表现出与红色幽灵行为完全相反的特性。实际上，平基会试图抢先出现在吃豆人前方，具体做法是将目标位置设定在吃豆人当前位置前方4格处。 即吃豆人移动的方向。不过有一个例外情况：当吃豆人向上移动时，由于代码中的缺陷，计算结果会包含一个与预期向上偏移量相等的向左偏移量。正是由于这个代码漏洞，计算过程中会加入一个等于预期向上偏移量的向左偏移量。	
利用吃豆人的预期位置，玩家可以操控平基来绕过导航代码的限制。这种情况在吃豆人与平基彼此距离极近时最为常见。例如，当两者距离非常接近时就会出现这类状况。 如果它们朝彼此移动，一旦间距小于4格，目标格子就会位于平基的后方。此时若平基遇到交叉路口，它会试图通过转向来折返，从而到达目标格子。处于这种状态时，一旦遇到交叉路口，平基就会尝试通过转弯折返，以便抵达目标格子。	
玩家们常常会通过迅速改变方向来施展所谓的“假动作”。当平基正接近一个交叉路口时。就在平基经过的瞬间移动目标瓷砖	
小型模型补全 当其向左移动时，它们就能得到“正确”的结果方向。 如果该位置实际上是距离更远的位置吃豆人原本所在的位置 由于吃豆人不会响应幽灵的移动，玩家的操作也无法影响它。此时幽灵会直接返回到玩家当前所在的位置。这是因为吃豆人本身并不会通过转向的方式来移动。通过转动。 那只幽灵必须前往另一个地点。只有回到起始位置，它才能完成这一过程，而实现方式多种多样。通常情况下，它只需直接离开该地点即可。例如，如果吃豆人从悬崖上跳下去，它也会回到消失前的那个地方。它会返回到消失前的位置，随后进入空间之中。对于这些幽灵而言，存在不少可能的行动路径。虽然存在其他可能的路径，但最显而易见的情形是：玩家希望避免在游戏中陷入困境。其目标很简单，只需让吃豆人持续留在游戏中即可。 时间长度与他之前所处的时间相当，约为1/3。 追逐。 在吃豆人的例子中，这种情况就类似于……在战斗中，吃豆人能够“迎头赶上”。	GPT-2文本生成功能 正要穿过十字路口时，吃豆人就会出现能够避开幽灵投射的攻击物并返回安全地带安全屋的位置。 巴斯特 黄色ghost的人工智能是专门为……设计的专为……而设计 玩家必须按下空格键才能打开安全屋的门，随后进入屋内寻找藏身在相邻房间黄色幽灵。 玩家必须按下空格键来显示保险箱。房子的门，然后进入门内寻找黄色幽灵躲在隔壁房间里。 玩家还需注意，幽灵是不会尝试离开的。如果它已经看到吃豆人在安全屋内，就不会试图离开那里。事实上，幽灵在最初阶段也不会尝试撤离。除非它们已经身处安全屋内，否则不会这么做。 这两种行为背后的原因在于，玩家并不总是会试图前往安全屋。玩家并非总会努力抵达安全屋。在出现吃豆人（或布林基）之前。 如果已经有幽灵（或其他玩家）身处安全屋，玩家也必须警惕这种故障现象。即当幽灵已经处于安全屋时。玩家便应设法在吃豆人或布林基离开后立即逃离。 一旦它们离开之后。此时，该幽灵就会试图离开安全屋。以与“追捕”模式相同的方式使用安全屋。 吃豆人 有一些幽灵会追踪吃豆人。这些幽灵are

表9。随机选取的未见上下文（顶部），以及来自最小规模模型（左侧）和最大规模模型（右侧）的未经筛选的补全结果。每个上下文的长度为768个标记，此处展示约256个标记对应的段落内容；补全结果的长度为256个标记，并会完整显示出来。生成过程中采用了基于 *k* = 40 的Top-*k* 随机采样方法。

Context (WebText test)	
Corporal Michael P. Goeldin was an unskilled laborer from Ireland when he enlisted in Company A in November 1860. Goldein survived the war. Corporal Patrick O’Neal, also from Ireland, first enlisted in 1854 and served with Company L, 3d U.S. Artillery, in Oregon. He returned to the East Coast and enlisted in the company in 1860. O’Neal served until 1874, when he was named superintendent of the National Cemetery at Willets Point, New York. Corporal Benjamin Browne was a shoemaker from Orange County, New York. In August 1862, he enlisted in the newly formed 124th New York Volunteers, and was one of sixty-one men who transferred into Company A that October. Browne reenlisted in the company in February 1864 while it was camped at Brandy Station. He returned to civilian life after completing his enlistment in 1867. On 10 June, Artificer William Collins was promoted to corporal, probably to fill a combat leadership void for the crossing of the James River. Collins’s service record does not reflect the qualities he demonstrated to earn this promotion, but he had obviously overcome some serious problems. Born in Sacketts Harbor, New York, Collins enlisted in the company in December 1853 at the age of twenty-two, and reenlisted in December 1858. Just a month before the war began in April 1861, Collins went ”over the hill” and was not caught until three years later. Returned to the company on 22 March 1864, he was tried	
Small model completion	GPT-2 completion
on military charges and was discharged from the U.S. Military. He returned to the company, and was assigned to Division of Internal Affairs during the Civil War. As an engineer, Collins also had training in engineering from his native New York. After graduate studies in the Union College of Engineering, he was employed in the United States Army for two years at the same time. There he worked under Captain Daniel Pfeiffer and was promoted to colonel in May 1870, three years after Captain Pfeiffer’s enlistment. Collins returned home to England after completing his first year in the First Division to serve the next two years in the Army Corps of Engineers, and by his second year, he began to experience his service duty.␣—endof␣text—␣MOSCOW, July 5 (Reuters) - U.K.-based Russian state media and government sources and experts say Moscow has sought to downplay the ”military buildup” of the Syrian opposition while maintaining support for President Donald Trump’s plan to build a wall along the U.S.-Mexico border, calling it a ”military coup”. U.S.-led Syria war veteran and Syria envoy Michael Flynn (R), right, gestures with U.S.-based Syrian political analyst Ahmet Koryat, with	in June of that year. Sentenced to be shot for desertion, he remained on active duty. Captain James J. Dall was born in Virginia in 1829 during the Revolution and joined his father in the Army in 1836. He served with the 2d Virginia Infantry until 1845, when he went to the Western Reserve Regiment; and, on his discharge, he reenlisted with Company I, 25th New York Volunteers (which in 1845 numbered about 4,500 men). One of his brothers, James, served with his company in the same capacity in the same brigade. While the regiment remained in Connecticut through 1846, Captain Dall served with Company I at various times before his discharge. A veteran of the Mexican War, he had served two years on the staff of Brigadier General John J. Sullivan at San Antonio. During the Peninsula Campaign of 1863, he commanded a company of twenty-three and was in charge of the battery of fifty-four heavy mortars and the gunboat Pinta of the Battery, a gunboat and several other boats. Captain Dall was on active duty with Company I in the fall of 1865. Two years later, he was promoted to second lieutenant. Captain John A. O’

Table 10. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top- k random sampling with $k = 40$ was used for generation.

上下文 (WebText测试)	
1860年11月，下士迈克尔·P·戈尔丁还是来自爱尔兰的一名非技术工人，他加入了A连。 戈尔丁在这场战争中幸存了下来。同样来自爱尔兰的下士帕特里克·奥尼尔则早在1854年就入伍，之后在俄勒冈州的美国第3炮兵部队的L连服役。之后他返回东海岸，并于1860年再次加入该部队。 奥尼尔一直服役到1874年，随后被任命为纽约威利茨角的国家公墓的主管。 本杰明·布朗下士是来自纽约州奥尔兰治县的制鞋匠。1862年8月，他加入了新组建的纽约第124志愿军，同时也是那批转调至A连的61名士兵之一。 这些士兵转调的时间恰好在10月。1864年2月，当A连驻扎在布兰迪站时，布朗再次加入该连队。 他在1867年完成服役后便回到了平民生活。 在1867年完成兵役后回归平民生活。 6月10日，工匠威廉·柯林斯被提升为下士，很可能是为了填补战斗指挥岗位的空缺。 因渡过詹姆斯河而获颁勋章。柯林斯的服役记录并未体现他在此过程中展现出的优秀品质。 他虽未因这些表现而获得晋升，但显然已克服了诸多严峻的挑战。他出生于纽约州的萨克茨港。 柯林斯在22岁时，即1853年12月，加入了第一连，并于1858年12月再次入伍。 在1861年4月战争爆发前一个月，柯林斯逃兵了，直到三年后才被抓获。 1864年3月22日，他重返第一连后便遭到了审判。	
小型模型完成	GPT-2文本生成功能
他曾因军事相关指控被美国军方解雇。 之后他重返该公司，并被分配了工作。 在内战期间，他被派往内部事务部门工作。 作为一名工程师，柯林斯还在其家乡纽约接受了工程方面的培训。完成研究生学业后他继续深造。 他在联合工程学院工作。 同一时期，他还在美国陆军服役了两年。 时间。在那里，他师从丹尼尔·菲弗上尉工作。 三年后，也就是1870年5月，他在菲弗上尉入伍之后被晋升为上校。 在普费弗船长入伍之后。柯林斯回来了 在陆军工程兵团完成第一年的服役后，他回到了英格兰。 他接下来两年将继续在陆军工程兵团服役。 到了第二年，他开始履行自己的兵役职责。 ——文末注释——莫斯科，7月5日（路透社）——总部位于英国的俄罗斯官方媒体…… 7月5日（路透社）——总部位于英国的俄罗斯官方媒体及…… 政府消息人士和专家称，莫斯科一直在寻求为淡化叙利亚反对派的“军事集结”态势 在维持对唐纳德总统的支持的同时调整立场 特朗普沿美墨边境修建隔离墙的计划 边界问题，并将其称为“军事政变”。 由美国牵头的叙利亚战争老兵及叙利亚问题特使迈克尔右边的弗林（共和党人）正在用位于美国的叙利亚政治人士示意。 分析师艾哈迈德·科里亚特，与	同年6月，他因擅离职守被判处枪决。 继续执行现役任务。 詹姆斯·J·达尔上尉于1829年出生在弗吉尼亚州，当时正值…… 美国独立战争期间，他于1836年随父亲加入了军队。他一直在弗吉尼亚州第二步兵团服役，直至1845年，那时他前往了西部预备团；随后，他 退伍后，他重新加入纽约第25团第一连。 志愿者（在1845年时约有4500名男性）。 他的兄弟之一詹姆斯曾随所在连队服役。 仍在同一旅中担任相同职务。尽管如此， 该团在1846年之前一直驻扎在康涅狄格州，达尔上尉在退役之前，曾多次在第一一连服役。 他的出院。作为美墨战争的退伍军人，他曾经历过那场战争。 他曾在美国圣安东尼奥为约翰·J·沙利文准将的部下服役两年。 在1863年的半岛战役期间，他负责指挥一个由23人组成的连队。 此外，他还掌管着一支由54门重型迫击炮构成的火力部队。 该部队还包括所属的炮艇“平塔号”以及其他几艘舰船。 达尔上尉当时正与这些部队一同执行作战任务。 几艘其他的船。达尔上尉当时正在执行任务。 1865年秋季的第一连。两年后，他便 晋升为少尉。 约翰·A·奥船长

表10. 随机选取的未见上下文（上方），以及来自最小规模模型（左侧）和最大规模模型（右侧）的未经过筛选的补全结果。每个上下文的长度为768个标记，其中展示了约256个标记对应的段落内容；补全结果的长度为256个标记，并会完整显示出来。生成过程中采用了基于 $k = 40$ 的Top- k 随机采样方法。

Language Models are Unsupervised Multitask Learners	
Context (WebText test) Xmonad, Xresources, and Fonts Xresources can be an absolute pain (they were for me). So, this a bit of help: First of all, the way that fonts are generally handled in Xmonad is through XFT . Xft is the X FreeType Interface library and is a library that handles the actual rendering of fonts in Xmonad/urxvt (Xresources specifies a bunch of resources to things that launch under the X window manager. Settings for individual programs are specified in Xresources like so: Urxvt.background:background settins here). Anyway, fonts are set using this syntax in Xresources : urxvt*font:xft:your font name goes here:size=the size of the font goes here,xft:fall-back font 1 goes here So, thats the most basic part of fonts in urxvt specified via Xresources. Fixing /.ssh/ Permissions in Cygwin In Cygwin on Windows I found that I could not set the permissions of my /.ssh/ folder to be 0600, as is required for ssh to allow you to use keys. The symptom I had was that I no matter what I did, it always modified the owner and the group	
Small model completion was set to /.ssh/ , which was the way things were. This could not be fixed completely. I found I could make that much more workable with just typing the username with your mouse then trying to set permissions to be ”0600” for the group, which is what my system is now configured to do. However, as of Cygwin 7 on Linux I have found that doing so is now ”unnecessary” for my system! I also installed a package called chroot. It does that by allowing you to install/upgrade to your system directory. So, as of Windows 8 and up, installing that can require chroot, so I tried and tried and found that it wasn’t so bad when installing the package on Linux and then uninstalling it. As of my last write-up on Cygwin, the script just does its job (the sudo commands do the rest), if you try it you will have to log the change. If that wasn’t confusing enough, this is what I tried and just went back to my /.ssh/ directory and the chroot script. I found I was finally running out of permissions (because i forgot to create a symlink to ssh from the file system and	GPT-2 completion , making these files read-only. I tried just starting ssh and using a program like passwd which modifies the owner/group to 0644 permissions, but it didn’t work. It was annoying but not nearly as bad as it could be, just annoying to have to fiddle with permissions. So I decided to do this as the reverse: For each .ssh folder in your Cygwin directory, set the permission of each file and directory to 2644. Then start ssh with: ssh-add /.ssh/ The first time you connect, your permissions of the /.ssh folder that you just had in Cygwin will be set to 0600. Then every time you connect again, your permissions will change to 0644. Makefile for Xresources as used in /.xsessionrc These are made in my .xsessionrc : .xprofile: .xsession-errors: color=”” For example: .xprofile: color=blue As the colors are stored in a separate file, the color in your /.xsession-errors will look like cyan instead of blue. This is mostly because, as of now, only the color is assigned to your

Table 11. Random unseen contexts (top), and non-cherry-picked completions from both the smallest (left) and largest (right) models. Contexts are 768 tokens, with approximately 256 tokens worth of paragraphs shown. Completions are 256 tokens and fully shown. Top-*k* random sampling with *k* = 40 was used for generation.

语言模型属于无监督多任务学习器。	
上下文（WebText测试）。 Xmonad , Xresources , 以及字体。 Xresources往往会让人大伤脑筋（对我来说就是如此）。因此, 这里有一些实用的建议: 首先, Xmonad 中处理字体的方式是通过 XFT。XFT 即为 X 环境下的 FreeType 接口库, 负责在 Xmonad 和 Urxvt 中实现字体的实际渲染功能（而 Xresources 则用于指定与 X窗口管理器相关的各类资源）。它正是用于处理字体渲染的接口库, 具体在 Xmonad 和 Urxvt 环境中承担字体渲染的任务（Xresources 用于定义在 X窗口管理器下启动的各种程序的相关配置）。这些配置是针对在 X窗口管理器下运行的各个程序而言的, 具体的设置内容会在 Xresources 文件中予以规定。在 Xresources 中, 相关设置如下所示: Urxvt.background: 此处用于指定背景设置。 无论如何, Xresources 中使用以下语法来设置字体。 urxvt*font:xft:此处填写所需字体名称:size=此处填写字体大小,xft:fall-back font 1 goes here 因此, 这就是通过Xresources在Urxvt中指定字体的最基础部分。 在Cygwin中修复/.ssh/目录的权限问题。 在Windows上的Cygwin环境中, 我发现无法将/.ssh/目录的权限设置为0600。而0600正是SSH允许使用密钥所必需的权限设置。无论我怎么做, 该环境总是会修改该目录的所有者和所属组。所有者与群组	
小型模型补全 被设置为/.ssh/目录, 那也是之前的处理方式。无法完全修复。我发现我可以解决那个问题。只需输入用户名即可, 操作起来方便得多然后使用鼠标尝试将权限设置为“0600”针对该组, 这正是我当前系统所配置的模式待处理。 不过, 在 Linux 环境下的 Cygwin 7 版本中, 我发现这样做对我的系统而言现已变得“没有必要”了! 此外, 我还安装了一个名为 chroot 的软件包。它通过允许用户将程序安装或升级到系统的目录中来实现这一功能。 因此, 从 Windows 8 及更高版本开始, 安装该软件包就又我需要使用 chroot 功能, 于是反复尝试, 结果发现它并不可用。在 Linux 上安装该软件包时情况糟糕至极, 之后……再将其卸载时也是如此。截至我上次撰写关于 Cygwin 的文章时, ……该脚本仅负责执行其预设任务（其余操作由 sudo 命令完成）, 如果……亲自试一试就会发现, 必须记录下所做的更改。 如果上述情况还不够令人困惑的话, 以下是我尝试过的方法……我刚回到我的/.ssh/目录以及对应的chroot脚本那里。我发现权限终于快用完了（原因是……）因为我忘了在文件系统中为ssh创建符号链接。	GPT-2文本生成功能 , 导致这些文件变成了只读状态。我试过直接启动ssh程序。还试过使用像passwd这样的程序来修改相关设置。我将所有者/群组权限设置为0644, 但该方法并未生效。虽然这很烦人, 但实际情况还没糟糕到那种地步, 仅此而已。不得不反复调整权限实在令人苦恼。 于是我决定反其道而行之: 在Cygwin目录下的每个`.ssh`文件夹中, 将其中所有文件和目录的权限都设置为2644。将每个文件和目录的权限设置为2644。然后通过以下命令启动SSH: ssh-add /.ssh/首次连接时, 您所使用的/.ssh目录的权限将被设置为0600。您在Cygwin中创建的/.ssh目录也会被赋予同样的权限设置。此后每次再次连接时, 该目录的权限都会自动更改为0644。用于.xsessionrc文件中的Xresources相关Makefile如下所示。用于/.xsessionrc 中的 Xresources 的 Makefile 这些内容是我在 .xsessionrc 文件中设置的: .xprofile: .xsession-errors: color=” ” 例如: .xprofile: color=blue 由于颜色信息存储在单独的文件中, 因此位于/.xsession-errors中的颜色会显示为青色而非蓝色。/.xsession-errors 文件中的文字将会显示为青色而非蓝色。 这主要是因为目前仅为颜色分配了对应值。已分配给您的

表11。随机选取的未见上下文（上方），以及来自最小规模模型（左侧）和最大规模模型（右侧）的非筛选性补全结果。这些上下文的总标记数为768，其中约256个标记对应的段落会被展示；而补全结果的长度为256个标记，会完整显示出来。生成过程中采用了基于*k* 随机排序的 *k* = 40 采样方法。

Language Models are Unsupervised Multitask Learners		
Context (WebText test) Aaron loves mint chocolate cake, but he requires that it be paired with mini chocolate chips, so I threw some of those in between the layers. I also had a few Peppermint Jo Jos on hand so I crushed them up and threw some of those in along with some crushed meringue cookies because, why not? It’s a total smorgasbord of minty chocolate chippy cookie crunchy goodness. I didn’t measure how much of each topping I used, but after I tasted the finished product, I wish I had added more. You can add anything you want- crushed candy canes, peppermint bark, etc. And don’t be afraid to use a heavy hand. Texture = good. If you don’t have 7-inch cake pans, you can get 3 shorter 8-inch layers out of this		
GPT-2 Completion 1 recipe. I made the longer version, then used the pan for a layer of buttercream frosting and stuck 1/3 of it into the fridge for a frosting topper. Just make sure you have all of this frosting prepared earlier than you make this cake and you won’t have to wait on it as your egg whites sit on the bottom of your mixing bowl. You want the egg whites whipped to stiff peaks? Then put your cake dish over a pot of simmering water and heat it up until it’s just barely warm. It doesn’t take more than 3-5 minutes; just enough to get it started	GPT-2 Completion 2 recipe. If you do, and you know how to make that cake, then your life will never be the same. I have two very simple rules when it comes to cookie crumbs or chocolate chips. 1) When I buy raw sugar, I do not use it until it’s cooled down and has at least half a teaspoon of sugar on it. That’s my rule. And 2) If I have a recipe that calls for something and I want to use it in a recipe, I do not substitute it with something else on-line. Like this. I found this really cute card with cute little kittens on	GPT-2 Completion 3 recipe, which will likely yield about the same amount of cake. Chocolate Buttermilk Cake Print Chunks of chocolate cake, coated in buttermilk and topped with crunchy chopped chocolate, drizzled with chocolate sauce. Simple but a nice change from chocolate cake! Author: Momma Made Ingredients For the chocolate buttermilk cake 1 1/2 cup unsalted butter , room temperature 1 1/2 cup granulated sugar 3 large eggs plus 1 egg yolk For the chocolate glaze 1/3 cup cocoa powder 1 3/4 cups powdered sugar 6 ounces semis
GPT-2 Completion 4 recipe. Meringue Cookies 1 3/4 cup butter, softened 1 cup sugar 1 egg yolk 3 T. heavy cream 3 1/2-4 cups flour pinch of salt Peppermint-Jojo Topping 1 1/4 cups powdered sugar 1 cup chopped pecans 1/2 cup finely chopped mint leaves 1/2 cup chopped fresh mint (about 1/2 sheet) 1 tsp. peppermint extract 1/4 tsp. sea salt pinch of sea	GPT-2 Completion 5 recipe. My 2 failures at Christmas morning were not due to poor frosting skills. That was another success story. The cake was beautifully moist, and well, just a beautiful cake. It’s got very little air in it or moisture, but all the chocolate and mint was perfectly blended together. One thing that made me cringe a bit was that all those ingredients-egg yolks, sugar, melted butter, chocolate chips, melted chocolate, and mint- had to be mixed right into the cake batter, which is super time-consuming. A friend suggested that if I put the batter in a separate bowl, I	GPT-2 Completion 6 recipe, which will give you about 6-9 servings of delicious chocolate cake. To prevent your layers from sticking together, I used my 8-inch round tart pan, lined it with parchment paper, then cut up 4 7-inch round cakes from it and sandwiched the layers between them (this also worked well for the 1-layer chocolate cake as well). When you’re ready to bake this cake, remove the parchment paper, preheat oven to 350 degrees and bake for about 30 – 40 minutes, until the top is golden and a tester comes out clean. Chocolate Buttermilk Layer Cake with Peppermint Jo

Table 12. Non-cherry-picked completions from GPT-2 generated from the same context (from WebText test). Context is 384 tokens (shown truncated), and generations are 128 tokens. Top- k random sampling with $k = 40$ was used for generation.

语言模型其实就是无监督多任务学习器		
上下文 (WebText测试) 亚伦非常喜欢巧克力蛋糕，但他要求里面必须加迷你巧克力豆，所以我加了一些进去。那些位于各层数层之间的部分。我手头还有些薄荷味乔乔糖，于是把它们捣碎后加入了一些进去。把那些东西再加上一些压碎的蛋白酥饼，反正也没什么不可以的。这简直就是薄荷巧克力风味的盛宴啊。酥脆的曲奇，口感十足。我没有精确计量每种配料的使用量，但在尝过成品之后这款产品，真希望我能加更多配料。你可以随心添加任何东西——比如压碎的糖果棒、薄荷皮等等。此外，不必顾虑用量过多，这样的处理方式能让口感 =更加出色。 如果你没有7英寸的蛋糕烤盘，用它也能做出3层较短的8英寸尺寸的蛋糕。		
GPT-2文本生成功能1 这是对应的食谱。我做了更长版本的版本，随后用这个烤盘来制作其中一层。黄油奶油糖霜有1/3粘在上面。把这部分放入冰箱，用来制作糖霜。还需要准备顶饰，务必确保所有材料都备齐。此糖霜是在更早之前准备好的只要提前做好这些，制作蛋糕时就无需再费心了。这样就不必像处理蛋白那样一直等待它凝固了。将其放在搅拌碗的底部。 碗。 你需要将蛋清打发至硬性发泡的状态，对吧？那么就把蛋糕模具放在一锅微沸的水上。把蛋糕放入该水中加热。持续加热，直到蛋糕温度刚好升至适宜的程度即可。温度适中。只需3到5分钟即可。这点时间刚好足以让它启动起来。	GPT-2文本生成功能2。 如果你们知道制作那种蛋糕的配方，那么你们的 lives 将从此不再相同。 生活也将因此发生彻底的改变。生活再也不会像从前那样了。 我有两条非常简单的规则。说到饼干碎的话…… 或是巧克力豆。1) 当我购买原糖时，我不会在它……完全冷却下来、表面至少有一茶匙糖分之前，不会使用它。这就是我的原则。 已经冷却，且温度至少降到了二分之一在上面加一茶匙糖。那就是我的。规则：2) 如果我有一份食谱，其中提到了某种食材，而我想在食谱中直接使用它，而不会用其他东西替代。比如就像这里所说的那样。 就是这样。 我觉得这真的很有用。可爱的卡片上面有可爱的小猫咪图案	GPT-2文本生成功能3 按照这个配方制作，大概能得到同等分量的蛋糕。 巧克力 脱脂牛奶 Cake 打印出巧克力蛋糕的小块，外层裹上脱脂牛奶并加以点缀。顶部撒上酥脆的巧克力碎。 细雨纷落 with 巧克力 酱汁。虽然简单，但却为常规口味带来了不错的变化。 巧克力蛋糕！制作人：妈妈。制作巧克力蛋糕所需的材料。脱脂牛奶 1又1/2杯（无盐）。黄油，室温状态。 1又1/2杯白砂糖 需要3个大鸡蛋加上1个蛋黄 制作巧克力糖浆需使用1/3杯可可粉 糖粉 1又3/4杯糖粉 6盎司半成品
GPT-2 Completion 4算法 食谱。 蛋白酥饼 1又3/4杯软化后的黄油 1杯糖 1个蛋黄 3汤匙浓奶油 3又1/2到4杯面粉 一小撮盐 薄荷乔乔糖霜 1又1/4杯糖粉 1杯切碎的山核桃 1/2杯切碎的薄荷叶 1/2杯切好的新鲜薄荷（约……） 1/2片 1茶匙薄荷提取物 1/4茶匙海盐 一小撮海盐	GPT-2文本生成功能5 配方。 我在圣诞节的两次失败经历那又是一个成功的案例。这块蛋糕的口感十分湿润可口。这块蛋糕湿润度极高，口感极佳。嗯，就是一块非常漂亮的蛋糕。它……里面的空气和水分都极少。不过所有的巧克力和薄荷成分都完美地融合在了一起。 有一件事让我感到有些尴尬，那就是所有那些配料——蛋黄、糖 融化的黄油 巧克力豆、融化的巧克力 还有薄荷——必须准确混合放入蛋糕中 面糊 也就是说极其耗时。 有一位朋友建议我把面糊放进另一个碗里，这样我	GPT-2文本生成功能6 这份食谱能让你做出约6到9份美味的巧克力蛋糕。为防止蛋糕的各层层间粘在一起，我使用了自己的8英寸圆形塔盘，并在其内衬了先铺上烘焙纸，然后将其剪成4块。再用这些材料制作出7英寸圆形蛋糕。将各层的蛋糕夹在烘焙纸之间。 这种方法对1层结构的巧克力蛋糕也同样适用。（同样的方法也适用于1层结构的巧克力蛋糕。） 准备好之后就可以开始烘烤了。取出蛋糕后，再取下烘焙纸。先将烤箱预热至350华氏度，然后开始烘烤。将蛋糕放入烤箱烘烤约30到40分钟。直到顶部呈现金色，然后由测试人员进行检查 如此即表示蛋糕已烤好。 巧克力 脱脂牛奶 层薄荷朱利安蛋糕

表12。这些补全结果并非从GPT-2中刻意筛选而出，而是基于相同的上下文（来自WebText测试）由该模型生成的。上下文长度为384个标记（此处仅显示部分内容），每次生成的文本长度为128个标记。在生成过程中采用了基于 k 个随机项的 $k = 40$ 顶级采样策略。

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.
GPT-2: The scientist named the population, after their distinctive horn, Ovid’s Unicorn. These four-horned, silver-white unicorns were previously unknown to science. Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved. Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow. Pérez and the others then ventured further into the valley. “By the time we reached the top of one peak, the water looked blue, with some crystals on top,” said Pérez. Pérez and his friends were astonished to see the unicorn herd. These creatures could be seen from the air without having to move too much to see them – they were so close they could touch their horns. While examining these bizarre creatures the scientists discovered that the creatures also spoke some fairly regular English. Pérez stated, “We can see, for example, that they have a common ‘language,’ something like a dialect or dialectic.” Dr. Pérez believes that the unicorns may have originated in Argentina, where the animals were believed to be descendants of a lost race of people who lived there before the arrival of humans in those parts of South America. While their origins are still unclear, some believe that perhaps the creatures were created when a human and a unicorn met each other in a time before human civilization. According to Pérez, “In South America, such incidents seem to be quite common.” However, Pérez also pointed out that it is likely that the only way of knowing for sure if unicorns are indeed the descendants of a lost alien race is through DNA. “But they seem to be able to communicate in English quite well, which I believe is a sign of evolution, or at least a change in social organization,” said the scientist.

Table 13. Conditional generation on an out-of-distribution context by GPT-2. Cherry pick of 10 samples generated with $k = 40$.

上下文（人类撰写）：一项令人震惊的发现表明，科学家们在安第斯山脉一个偏远且此前未被探索过的山谷中，发现了一群独角兽栖息于此。更让研究人员惊讶的是，这些独角兽竟然能说流利的英语。让研究人员感到惊讶的是，那些独角兽居然能说流利的英语。 GPT-2：这位科学家根据它们独特的角形特征，将这群独角兽命名为奥维德的独角兽。这类拥有四只角、呈银白色的独角兽在科学界此前是完全未知的存在。 时至今日，近两个世纪过去后，引发这一奇特现象的谜团终于被解开了。最终得以解决。 Dr. 来自拉巴斯大学的进化生物学家豪尔赫·P·埃雷斯与几位同伴在探索安第斯山脉时，发现了一个没有其他动物或人类的小山谷。那些伙伴们在探索安第斯山脉时，发现了一个没有其他动物的小山谷。P·埃雷斯注意到，这个山谷里似乎有一处天然喷泉，周围环绕着两座岩石峰顶与银装素裹的雪坡。两座由岩石与银色积雪构成的山峰。 佩雷斯和其他人随后继续向山谷深处前进。 “等到我们抵达山顶时， ”佩雷斯说道： “那处的湖水呈蓝色，水面还漂浮着一些水晶。” 佩雷斯和他的朋友们看到那群独角兽时大为震惊。 这些生物很可能就是……从空中就能看到它们，而且几乎近在咫尺，仿佛一伸手就能触碰到。它们的角。 在研究这些奇特的生物时，科学家们发现这些生物还具有其他特点。他们还能说一些相当标准的英语。佩雷斯指出： ‘例如，我们可以看出它们拥有某种共同的’语言‘，类似某种方言或变体。’ Dr. 佩雷斯认为，独角兽可能起源于阿根廷，因为在当地人们曾相信这些动物是早在人类到来之前就生活在那里的一个失落种族的后代。人们认为它们是人类到来之前居住在那里的某个失落种族的后代。在南美洲的那些地区。 尽管它们的起源仍不明确，但也有人认为这些生物可能是被创造出来的。在人类文明出现之前的某个时代，当一位人类与一只独角兽相遇时。据P´erez所述，在南美洲，这类事件似乎相当常见。 不过，P´erez也指出，要想确定这一点，唯一的办法很可能是通过DNA可以证实，独角兽确实是某个已消失的外星种族的后代。 “但它们似乎还具备……”我的英语沟通能力相当不错，我认为这既是进化的体现，至少也代表着某种变化。 ‘这就是社会结构的体现。’ 那位科学家如此说道。
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表13. GPT-2在非典型上下文中的条件生成情况。这些样本是从使用 $k = 40$ 生成的10个结果中精选出来的。

Language Models are Unsupervised Multitask Learners	
Article: Prehistoric man sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave in modern day France 36,000 years ago. Now, with the help of cutting-edge technology, those works of art in the Chauvet-Pont-d’Arc Cave have been reproduced to create the biggest replica cave in the world. The manmade cavern named the Caverne du Pont-d’Arc has been built a few miles from the original site in Vallon-Pont-D’arc in Southern France and contains 1,000 painstakingly-reproduced drawings as well as around 450 bones and other features... Cavemen and women sketched an incredible array of prehistoric beasts on the rough limestone walls of a cave 36,000 years ago and now a replica has been created (pictured) ...	
GPT-2: The original site in Vallon-Pont-D’arc in Southern France is a Unesco World Heritage site and is the oldest known and the best preserved cave decorated by man. The replica cave was built a few miles from the original site in Vallon-Pont-D’Arc in Southern France. The cave contains images of 14 different species of animals including woolly rhinoceros, mammoths, and big cats.	Reference: Cave mimics famous Caverne du Pont-d’Arc in France, the oldest cave decorated by man and the best preserved. The replica contains all 1,000 paintings which include 425 such as a woolly rhinoceros and mammoths. Minute details were copied using 3D modelling and anamorphic techniques, often used to shoot widescreen images. The modern cave also includes replica paw prints of bears, bones and details preserved in the original cave.
Article: ‘If you can find my submarine, it’s yours,’ Russian oil billionaire Roman Abramovich once said. And it seems the oligarch and Chelsea FC owner, whose submarine is just one of the extras that came with his £300million superyacht Eclipse (perfect for getting to shore undetected), is not the only wealthy businessman splashing out on underwater exploration. Dubbed Earth’s real ‘final frontier’, the oceans are still so little-explored that billionaires are queuing up to buy vessels that give them a glimpse of the dramatic seascapes and incredible wildlife of the world’s oceans. So if you have a spare few million in the bank and want some holiday snaps and Instagram posts that will really trump everyone else, you may want to snap up one of these... Whale of a time: The OrcaSub takes you 2000 feet beneath the surface in two pressurised Perspex viewing domes for optimum exploration ...	
GPT-2: The U-Boat Worx submarine can dive to 984 feet (300 metres) and is capable of diving to the bottom of the ocean. It can be used for diving, snorkelling and even snorkel diving. The submarine is also equipped with a snorkelling platform.	Reference: Feel like a James Bond villain with the most advanced submerged ocean vehicles. Some convert from a yacht into a submarine, and others detached off for a speedy exploration. Oliver’s Travels offers a Mile Low package on its submarine, complete with chef and butler for £175,000 a night.
Article: Amina Ali Qassim is sitting with her youngest grandchild on her lap, wiping away tears with her headscarf. Only a few months old, this is the baby girl whose ears she desperately tried to cover the night the aerial bombardment started. She lay awake, she says, in a village mosque on the Yemeni island of Birim, counting explosions as the baby cried. It could have been worse though. They could have still been in their house when the first missile landed. ”Our neighbor shouted to my husband ’you have to leave, they’re coming.’ And we just ran. As soon as we left the house, the first missile fell right by it and then a second on it. It burned everything to the ground,” Qassim tells us ...	
GPT-2: Yemen is in the middle of a civil war. Saudi Arabia is leading the coalition bombing campaign. It’s been bombing Yemen for more than two months now.	Reference: Amina Ali Qassim’s family sought shelter in a mosque before fleeing Yemen. Thousands like them are boarding boats to sail to Djibouti. Saudi Arabia has been pounding Yemen in a bid to defeat Houthi rebels.

Table 14. Summaries generated by GPT-2 and reference summaries on the test set of the CNN and Daily Mail dataset.

语言模型属于无监督多任务学习器。	
文章标题: 36000年前，史前人类在现代法国一处洞穴的粗糙石灰岩壁上绘制了大量惊人的史前野兽图像。36,000年前的现代法国。如今，在先进技术的助力下，肖维-蓬达尔克洞穴中的那些艺术作品已被复原。由此打造出了世界上规模最大的洞穴复制品。这座名为蓬达尔克洞窟的人造洞穴建在距离原始遗址几英里之外的地方。位于法国南部的瓦隆-蓬达尔克拥有1000幅精心复制的绘画作品以及大约450件其他物品。骨骼及其他特征 ... 远在36000年前，穴居人就在洞穴粗糙的石灰岩壁上绘制了大量史前动物的形象。如今，人们已经根据这些古老遗迹制作出了复制品（如图所示）。 ...	
GPT-2: 位于法国埃尔恩地区的原始遗址是联合国教科文组织世界遗产地。它正是人们所熟知的那个著名遗址。已知最古老且保存最为完好的、由人类装饰的洞窟。这座复原洞窟建在法国南部瓦隆-蓬达尔克地区的原始遗址数英里之外。该洞窟内绘有14种不同动物的图像。其中包括长毛犀、猛犸象以及大型猫科动物。<style id='1'>参考资料: </style>该洞窟模仿的是著名的蓬达尔克洞窟。	参考来源: 该洞穴模仿了著名的蓬达尔克洞窟在法国，存在最古老的由人类绘制的洞穴。保存最为完好。 该复制品完整再现了全部1,000幅壁画。其中包含425种不同的动物形象，例如长毛犀和猛犸象。通过三维建模以及常用于电影拍摄的变形技术，这些细微的细节都得到了精准还原。叙事与变形技术，常用于拍摄宽屏图片。这座现代化的洞穴内还设有复制品。熊的爪印、骨骼以及各种细节都保存在原洞穴中。即原始的洞穴。
文章内容: 俄罗斯石油大亨罗曼·阿布拉莫维奇曾说过：‘如果你能找到我的潜艇，它就是你的了。’而这位寡头兼切尔西足球俱乐部的老板，他的潜艇只不过是其所拥有的众多奢华物品之一而已。£价值3亿英镑的超级游艇“日食号”（非常适合在不被发现的情况下靠岸）并非唯一拥有此类奢侈资产的富商。大笔投资于水下探索事业。作为地球真正的“最后边疆”，海洋至今仍几乎未被充分探索，以至于许多亿万富翁纷纷排队想要购买能够让他们一探究竟的装备。这些装备能让他们亲眼目睹世界各大海洋中壮丽的海景以及令人惊叹的野生动物。因此，如果您银行里有多余的几百万美元，又希望拍出一些出色的度假照片并发布在Instagram上，从而真正让其他人自愧不如，那么您或许可以考虑购置其中一款……极致的潜水体验：OrcaSub通过两个加压的有机玻璃观察舱，将您带往2000英尺深的海底。为实现最佳探索效果。 ...	
GPT-2: U-Boat Worx潜水器能够下潜至984英尺深处（约合300米），并可抵达海洋底部。它可用于潜水、浮潜，甚至更多用途。该潜水器还配备有用于浮潜的平台。此外，它还具备浮潜功能。即专门用于浮潜的平台。	参考: 拥有最先进的潜水海洋载具，让你仿佛成了詹姆斯·邦德笔下的反派人物。先进的深海载人潜水器。有些可从中转换而来《奥利弗的旅行》推出了一项名为“Mile Low”的特色套餐，该套餐可在潜艇上享受服务。这套套餐包含厨师和管家，价格为每晚<style id='1'>£</style>175,000。配备厨师和管家的全套海底潜艇套房 每晚£175,000。
文章标题: 阿米娜·阿里·卡西姆正让最小的孙辈坐在自己腿上，用头巾擦去孩子的眼泪。这名女婴仅几个月大，就在空袭开始的那个夜晚，她曾拼命试图遮住自己的耳朵。据她所说，当时她躺在也门比里姆岛上一座乡村清真寺里，一边数着爆炸声，一边等待着婴儿的哭声响起。宝宝哭了。不过情况本可能更糟——当第一枚导弹击中时，他们仍有可能待在自家房子里。'我们的邻居对我丈夫大喊：“你们必须立刻离开，他们要来了！”于是我们就立刻逃走了。一离开那里……'“房子旁边，第一枚导弹直接击中了它附近，紧接着第二枚也落在了上面。一切都烧成了灰烬，”卡西姆向我们讲述道。 ...	
GPT-2: 也门正处于内战之中。沙特阿拉伯阿拉伯半岛正在引领这场联军轰炸行动。它就是已经对也门进行轰炸两个多月了。	参考来源: 阿米娜·阿里·卡西姆的家人寻求庇护于逃离也门之前的一座清真寺。像他们这样的数千人他们正登船前往吉布提。沙特阿拉伯一直在对也门发动猛烈攻击，意图击败胡塞叛军。对也门发动猛烈攻势，旨在击败胡塞叛军。

表14。由GPT-2生成的摘要以及CNN与《每日邮报》数据集测试集上的参考摘要。

English reference	GPT-2 French translation
This re-release, titled The Next Day Extra, was presented in the form of three disks: the original album, unpublished studio sessions and remixes, plus a DVD containing the four clips that have already been unveiled.	Les nouvelles re-releases, tout en premier disc, nécessaire de l'album, un studio session et remixes, plus une DVD de l'écran de quelques clips qui ont été déjà échappés.
French reference	GPT-2 English translation
Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu'il avait reconnu avant de faire, le fonctionnement de la hernia fonctionnelle que j'ai réussi, j'ai réussi.	This release, originally released on three discs: the original album, sessions studio inédites and remixes, a DVD containing the four clips that were released on the previous day.
English reference	GPT-2 French translation
One man explained that the free hernia surgery he'd received will allow him to work again.	Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu'il avait reconnu avant de faire, le fonctionnement de la hernia fonctionnelle que j'ai réussi, j'ai réussi.
French reference	GPT-2 English translation
Un homme a expliqué que l'opération gratuite qu'il avait subie pour soigner une hernie lui permettrait de travailler à nouveau.	A man told me that the operation gratuity he had been promised would not allow him to travel.
English reference	GPT-2 French translation
Speaking to an open government conference in London via video link, Mr Kerry said: "There is no question that the President and I and others in government have actually learned of some things that had been happening on an automatic pilot because the ability has been there, going back to World War Two and to the very difficult years of the Cold War, and then, of course, 9/11."	Le premier conférence de l'édition internationale en ligne sur la vide, le ministre de l'Édition internationale a été vide, lui dit : "Il n'y a qu'une question que le président et moi et tous les autres personnes d'une administration qui a été évident par la fois, par la part de la guerre, par la part de la guer
French reference	GPT-2 English translation
S'exprimant lors d'une conférence intergouvernementale à Londres par liaison vidéo, M. Kerry a déclaré: "Il est indéniable que le Président, moi-même et d'autres membres du gouvernement avons pris connaissance de certaines choses en mode pilote automatique parce que nous en avons la possibilité, dès la Seconde guerre mondiale et jusqu'aux années difficiles de la Guerre froide, puis bien sûr le 11 septembre."	In a conférence between the United States and London, Secretary of State John Kerry said: "It is undeniable that the President, myself and others of the government have been aware of certain certain choices that have been made in the past in order to be able to do certain things in a more automated way."

Table 15. English to French and French to English translations generated by GPT-2.

英文参考资料。 这款名为《The Next Day Extra》的重新发行作品以三种光盘的形式呈现。其中包括未公开发行的原版专辑、录音室录音及混音曲目，此外还有一张包含相关内容的DVD光盘。 录音室录音与混音曲目，再加上一张包含相关内容的DVD光盘已有四段视频片段被公开亮相。	GPT-2法文翻译版本。 作为首张光盘，这些新版重新发行版中包含了专辑的录音室版本、混音曲目，还有一张DVD光盘。 此外，还有几段视频片段的屏幕截图，展示了它们是如何被制作出来的。从‘屏幕上可以看到一些片段，它们‘且‘以‘方式’展示了‘种种奇观。
法语参考资料 有位男士正在解释功能性疝气的运作机制。他在开始阐述之前，先确认了自己所了解的功能性疝气的运作方式。我也曾研究过功能性疝气的运作机制，对此也有自己的理解。也是如此。	GPT-2的英语翻译任务 此版本最初以三张光盘的形式发行：原版一张专辑、在‘制作的录音室混音曲目，以及一张DVD光盘其中包含了之前发布的四段视频片段。day.
英语参考资料。 有位男士解释说，他即将接受的免费疝气手术将使他能够重新工作。康复后他就能重新工作了。	GPT-2完成的法语翻译版本。 有位男士在说明疝气手术的运作原理。他之前就认识的那项功能性功能——我也曾设想过针对功能性疝气的处理方法，我已也曾设想过。
法语参考资料 有位人士解释说，他曾经享受过的那次免费手术接受手术治疗疝气后，他便能重新工作。再次开始工作。	GPT-2英文翻译结果 有位男士告诉我，他原本应获得的手术费用补助并未让他有机会外出旅行。
英文参考资料 在伦敦举行的政务公开会议上发表讲话通过视频链接，克里先生表示：“毫无疑问”总统、我以及政府中的其他人士实际上得知了一些此前正在发生的事情自动驾驶模式，因为这种能力早已存在，随时可启动。回到第二次世界大战以及那些极其艰难的岁月冷战，接着是，当然，9-11事件。	GPT-2法语翻译 ‘国际在线版的首届会议关于真空的会议，由……部长主持 国际版共有‘种‘种语言版本。他看着对方说道：‘总统只有一个问题。’以及我与政府里的其他所有人。这一情况因战争而变得极为明显。战争的缘故。
法语参考资料。 他在一次政府间会议上发言时‘通过视频连线身处伦敦的克里先生表示：‘这是’‘总统、我本人以及政府的其他成员都无可否认地已经意识到了某些事情’‘我们一直以自动模式处理着这些事务，因为’‘我们……’ 在第二次世界大战期间，飞机便具备了‘这种能力，直至冷战时期的艰难岁月，这种情况依然存在。也就是在9月11日那次事件之后。	GPT-2的英语翻译任务 在美国与伦敦之间举行的一场会议上，国务卿约翰·克里表示：‘毫无疑问，’总统、我本人以及政府中的其他成员都曾……’早已知晓此前所做出的某些选择。过去人们正是为了能够以更为自动化的方式完成某些任务。以自动化方式。

表15. 由GPT-2生成的英语到法语以及法语到英语的翻译任务。

Context (passage and previous question/answer pairs) The 2008 Summer Olympics torch relay was run from March 24 until August 8, 2008, prior to the 2008 Summer Olympics, with the theme of “one world, one dream”. Plans for the relay were announced on April 26, 2007, in Beijing, China. The relay, also called by the organizers as the “Journey of Harmony”, lasted 129 days and carried the torch 137,000 km (85,000 mi) – the longest distance of any Olympic torch relay since the tradition was started ahead of the 1936 Summer Olympics. After being lit at the birthplace of the Olympic Games in Olympia, Greece on March 24, the torch traveled to the Panathinaiko Stadium in Athens, and then to Beijing, arriving on March 31. From Beijing, the torch was following a route passing through six continents. The torch has visited cities along the Silk Road, symbolizing ancient links between China and the rest of the world. The relay also included an ascent with the flame to the top of Mount Everest on the border of Nepal and Tibet, China from the Chinese side, which was closed specially for the event. Q: What was the theme A: “one world, one dream”. Q: What was the length of the race? A: 137,000 km Q: Was it larger than previous ones? A: No Q: Where did the race begin? A: Olympia, Greece Q: Is there anything notable about that place? A: birthplace of Olympic Games Q: Where did they go after? A: Athens Q: How many days was the race? A: seven Q: Did they visit any notable landmarks? A: Panathinaiko Stadium Q: And did they climb any mountains? A: Model answer: Everest Turker answers: unknown, yes, Yes, yes
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Table 16. Selected CoQA completion.

上下文（段落以及之前的问答对） 2008年夏季奥运会火炬接力在2008年3月24日开始，至2008年8月8日该届奥运会开幕前结束，其主题为“同一个世界，同一个梦想”。有关此次火炬接力的计划于2007年4月26日在中国北京公布。主办方将这次接力活动称为“和谐之旅”，整个接力过程持续了129天，火炬传递距离达到了137,000公里（85,000英里），这一距离是自1936年夏季奥运会开始这一传统以来的最长纪录。 3月24日，火炬在希腊奥林匹亚的奥林匹克运动会发源地点燃后，先后抵达了雅典的帕纳辛奈科体育场，随后于3月31日到达北京。从北京出发后，火炬沿着途经六大洲的路线继续前行。它还造访了丝绸之路沿线的诸多城市，象征着中国与世界其他地区古老的联系。此外，火炬传递还包含了一次特殊的登顶活动——火炬被带到了位于中国西藏与尼泊尔边境的珠穆朗玛峰顶，为此次活动，该区域特意关闭了普通游客的进入通道。 问：主题是什么？ 答：“同一个世界”，“同一个梦想”。 问：这场比赛的距离是多少？ 答：137,000公里 问：它的长度比之前的比赛更长吗？ 答：不是。 问：比赛是在哪里开始的？ A：希腊的奥林匹亚。 Q：那个地方有什么值得注意的地方吗？ A：它是奥林匹克运动会的发源地。 Q：之后比赛又移到了哪里？ A：雅典。 Q：比赛持续了多少天？ A：七。 Q：他们有参观过任何著名的地标吗？ A：帕纳辛奈科体育场。 Q：那他们有去爬山吗？ A: 模型回答：珠穆朗玛峰。 Turker的回答：未知、是、是、是。
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表16.精选的CoQA补全结果。

Context (passage and previous question/answer pairs) Tom goes everywhere with Catherine Green, a 54-year-old secretary. He moves around her office at work and goes shopping with her. ”Most people don’t seem to mind Tom,” says Catherine, who thinks he is wonderful. ”He’s my fourth child,” she says. She may think of him and treat him that way as her son. He moves around buying his food, paying his health bills and his taxes, but in fact Tom is a dog. Catherine and Tom live in Sweden, a country where everyone is expected to lead an orderly life according to rules laid down by the government, which also provides a high level of care for its people. This level of care costs money. People in Sweden pay taxes on everything, so aren’t surprised to find that owning a dog means more taxes. Some people are paying as much as 500 Swedish kronor in taxes a year for the right to keep their dog, which is spent by the government on dog hospitals and sometimes medical treatment for a dog that falls ill. However, most such treatment is expensive, so owners often decide to offer health and even life _ for their dog. In Sweden dog owners must pay for any damage their dog does. A Swedish Kennel Club official explains what this means: if your dog runs out on the road and gets hit by a passing car, you, as the owner, have to pay for any damage done to the car, even if your dog has been killed in the accident. Q: How old is Catherine? A: 54 Q: where does she live? A: Model answer: Stockholm Turker answers: Sweden, Sweden, in Sweden, Sweden

Table 17. Selected CoQA completion.

上下文（段落及之前的问答对） 汤姆始终跟在54岁的秘书凯瑟琳·格林身边，无论她走到哪里都会跟着。他在她的办公场所里来回走动，还会和她一起购物。凯瑟琳认为汤姆非常可爱，她说：‘大多数人似乎都不介意有汤姆在。’她还表示汤姆就像是她的第四个孩子，因此才会这样看待并照顾他。虽然汤姆会自己去买食物、支付医药费和税款，但实际上它只是一只dog而已。 凯瑟琳和汤姆居住在瑞典，这个国家要求每个人都必须按照政府制定的规则过有序的生活，同时政府也会为民众提供高水平的保障服务。而这类保障服务是需要投入资金的。 在瑞典，人们需要对所有物品缴税，因此他们并不惊讶养狗会带来更多的税负。有些人每年甚至要为养狗缴纳高达500瑞典克朗的税款，这些钱会被政府用于建设狗类医院，或是为生病的狗提供医疗救治。不过大部分此类医疗治疗费用都非常高昂，所以狗主人往往不得不为它们的健康乃至生命付出代价。 在瑞典，狗主人必须为自己宠物造成的任何损害负责。一位瑞典养犬俱乐部的工作人员对此做了说明：如果你的狗跑到马路上被过往车辆撞到，作为主人的你必须承担对车辆造成的所有损失，即便你的狗在事故中已经死亡。 问：凯瑟琳多大了？ 答： 54岁。 问：她住在哪里？ A: 模型回答：：斯德哥尔摩 Turker回答：：瑞典，瑞典，在瑞典，瑞典

表17。所选的CoQA补全结果。